

SATELLITE IMAGE CLUSTERING USING POSSIBILISTIC C-MEANS (PCM) ALGORITHM

GNR602: PROGRAMMING ASSIGNMENT

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PROBLEM STATEMENT

Cluster a satellite image using Possibilistic C-Means (PCM) algorithm and evaluate the cluster quality for different values of 'C' (at least 5 different values of C).

CONCEPT

- Possibilistic C-Means or PCM is a clustering algorithm derived from the fuzzy C-means clustering algorithm, the main difference being the relaxation of the constraint that the sum of the memberships is 1.
- This allows the memberships to be absolute and not relative like FCM, which means that they can be interpreted as the degree of compatibility with a cluster.
- The membership values and cluster centres are updated in each iteration according to the expressions listed below.
- After processing, the output clustered image is displayed along with the quality metrics for the different values of C.

CONCEPT

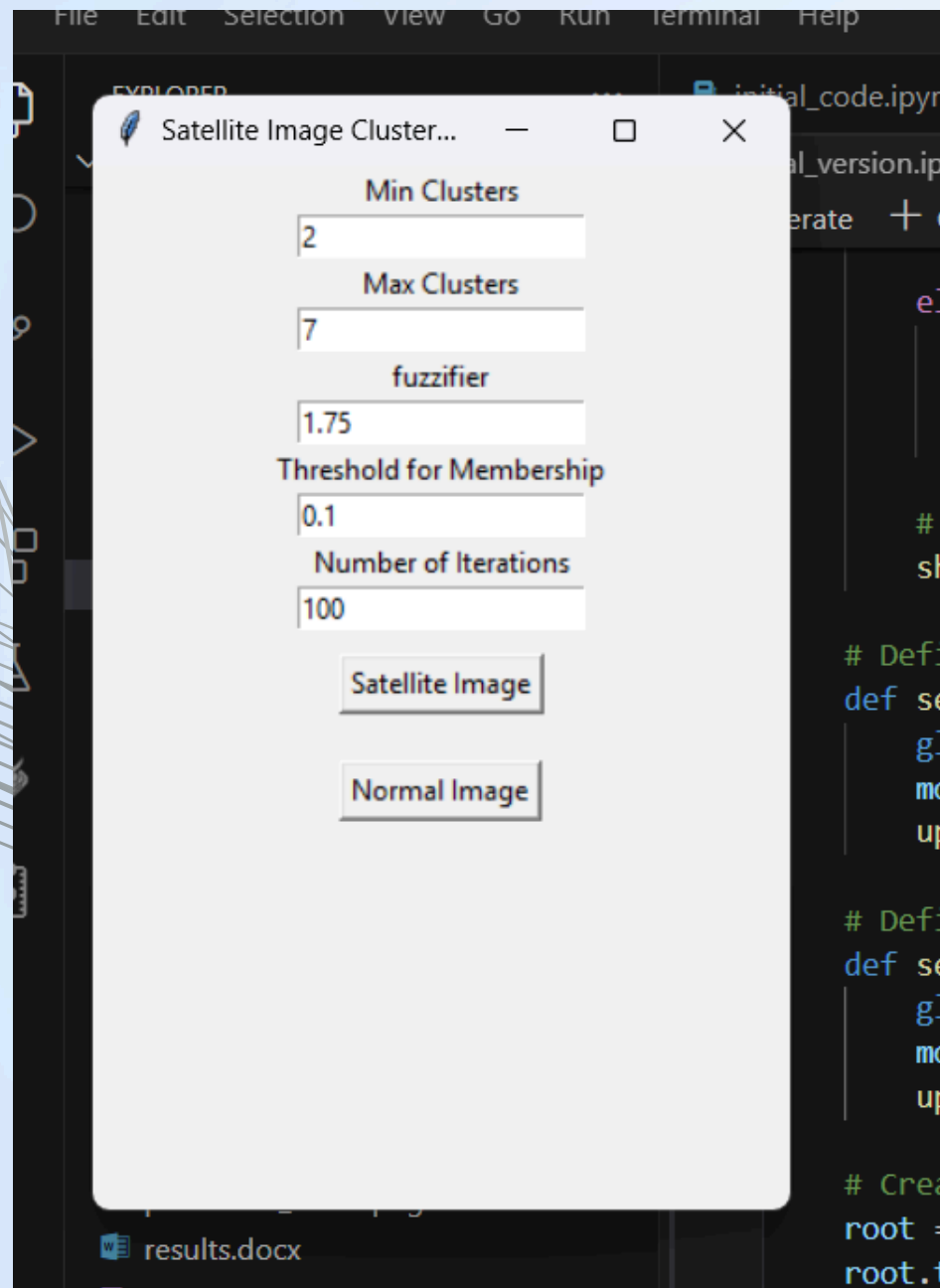
$$u_{ij} = \frac{1}{1 + \left(\frac{d_{ij}^2}{\eta_i} \right)^{\frac{1}{m-1}}}.$$

$$\mathbf{c}_i = \frac{\sum_{j=1}^N u_{ij}^m \mathbf{x}_j}{\sum_{j=1}^N u_{ij}^m}.$$

APPROACH

- Data Preparation: From a given satellite image, we extract the following bands, Red, Blue, Green, IR, NDVI, NDWI.
- The reasoning for this is that RGB gives most of the visual information about the image of how it appears to us, and IR, NDVI, and NDWI give information about the vegetation and water boundaries in the image.
- The last 3 bands are obtained if the false color image is also given to the program.
- Evaluation: The following quality metrics were chosen,
 - Simplified Silhouette Score
 - Davies-Bouldin Index
 - Dunn Index
 - RMSE
 - Xie-Beni Index

IMPLEMENTATION



User interface has the following parameters which need to be entered by the user :

Minimum number of clusters

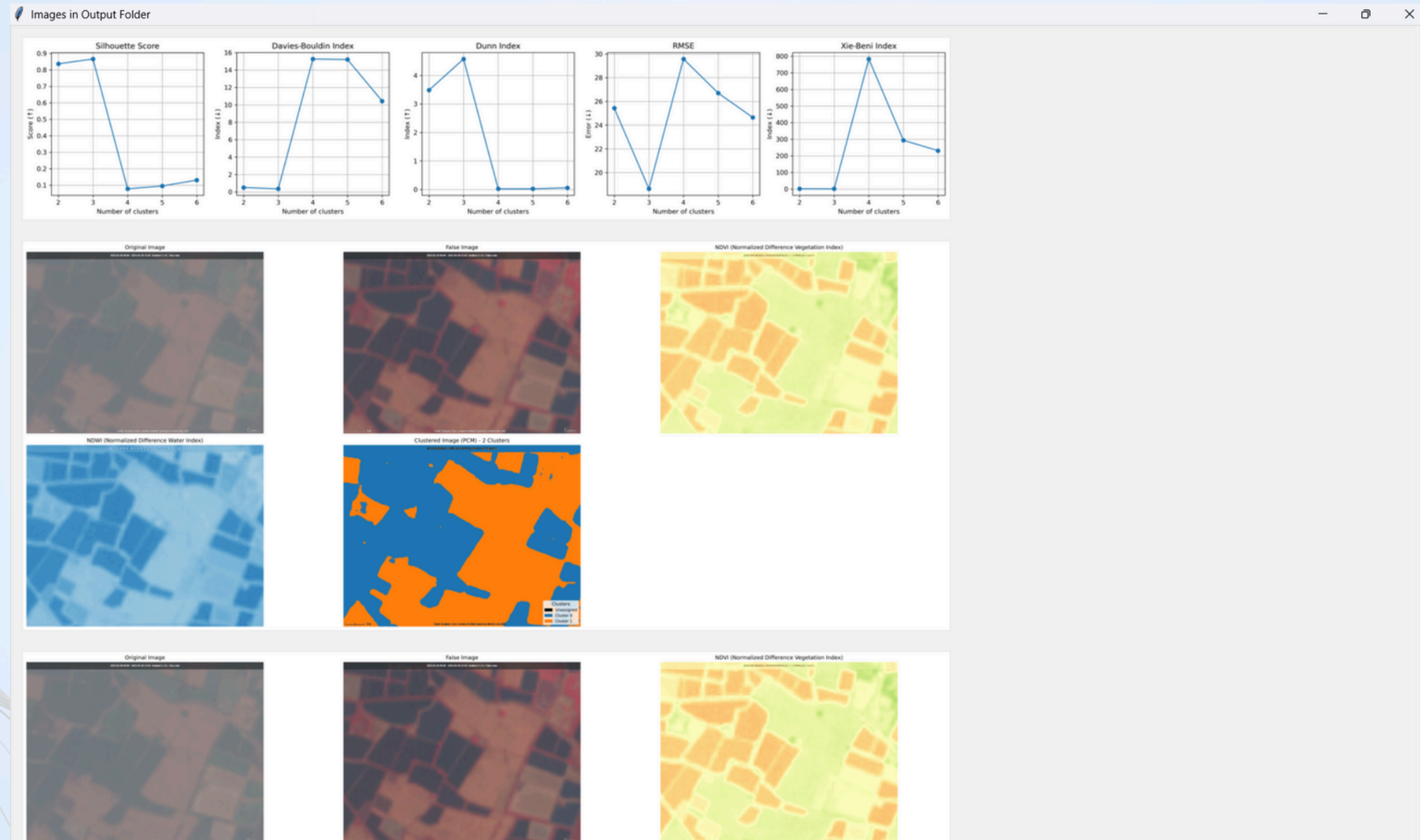
Maximum number of clusters

Threshold above which a point will be considered to be a part of any cluster

Maximum number of iterations

The user can choose to either cluster a satellite image or a non-satellite image. Based on the choice, the user is asked to choose files and an output folder to store the results in.

IMPLEMENTATION

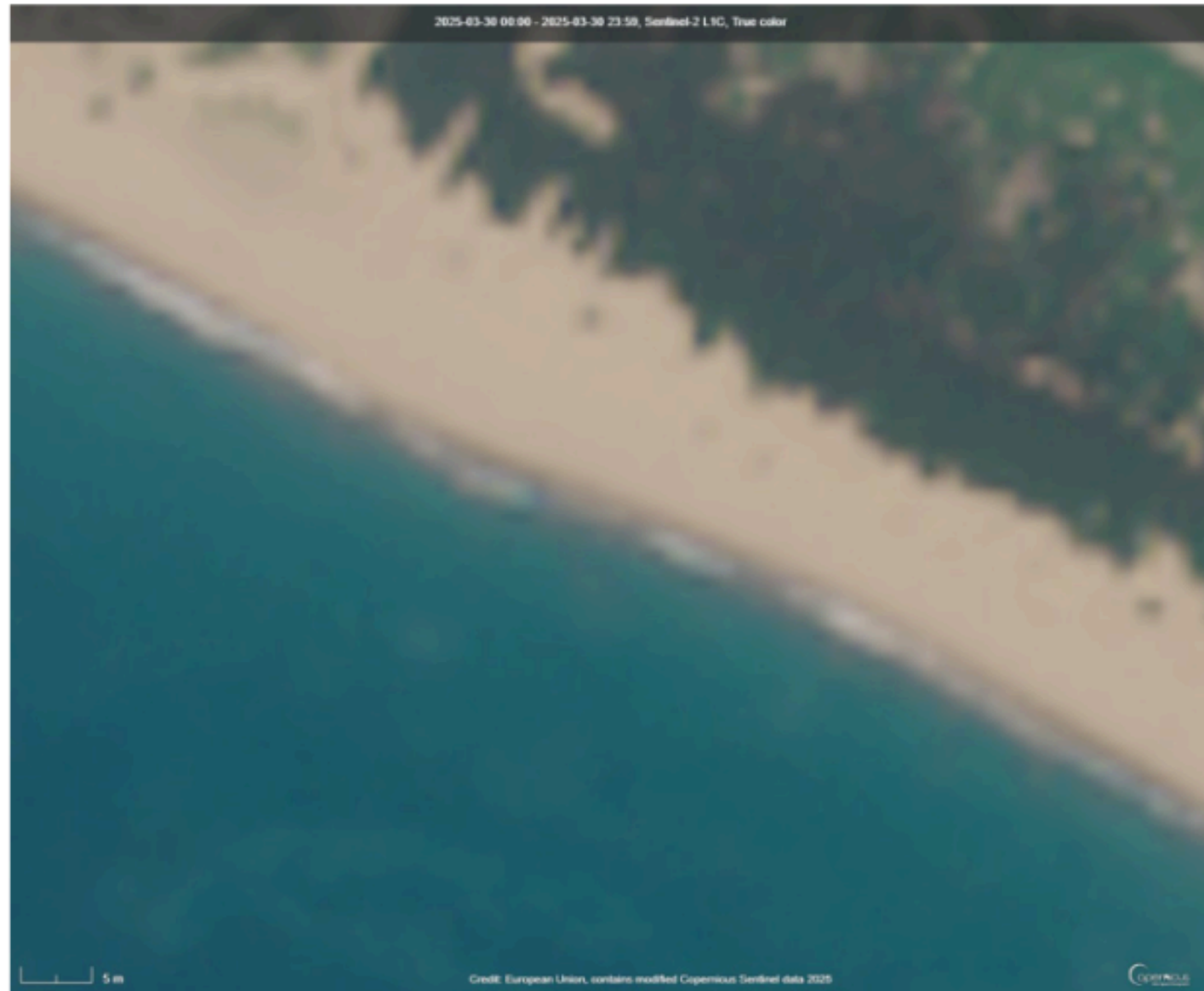


Once the algorithm runs fully, the results are displayed in a separate window

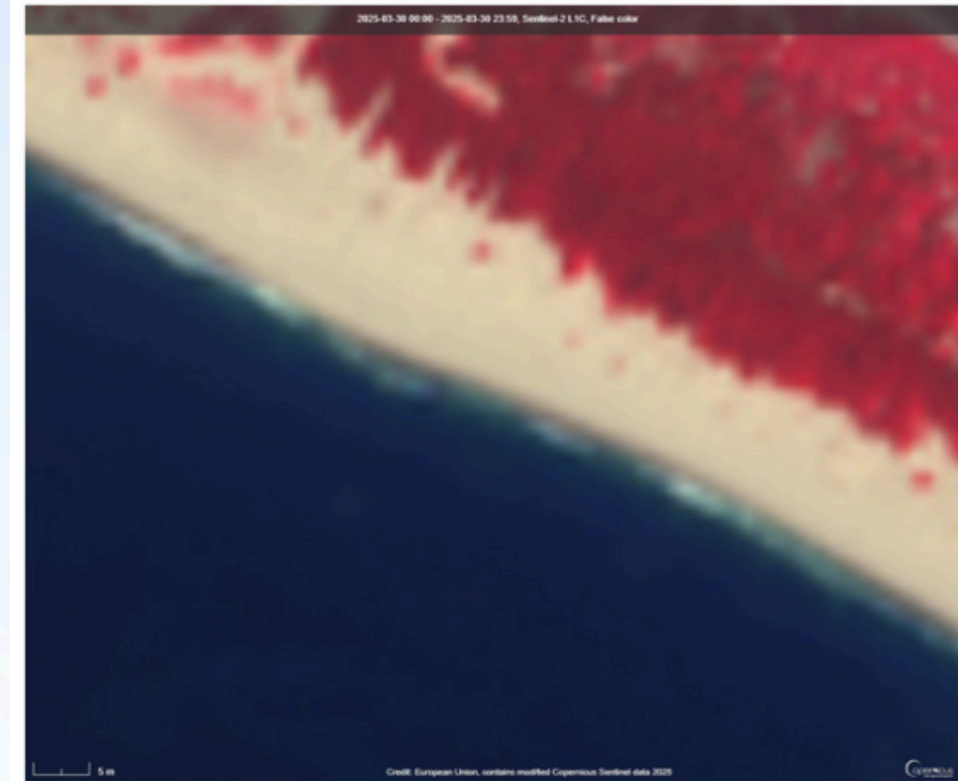
RESULTS

EXAMPLE - 1

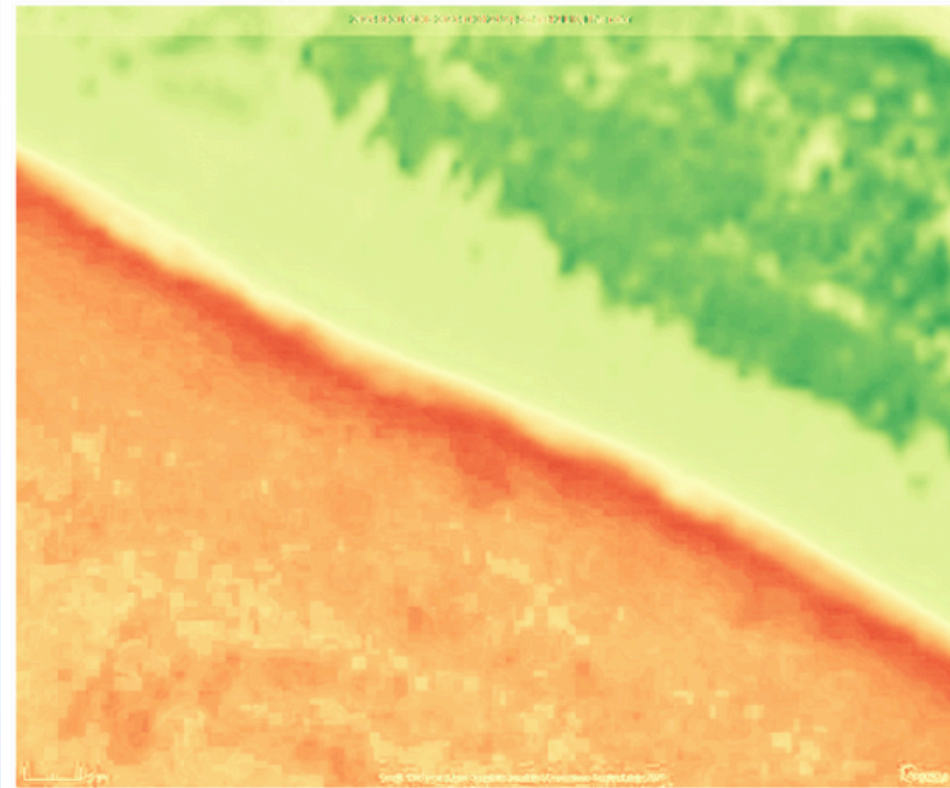
Original Image



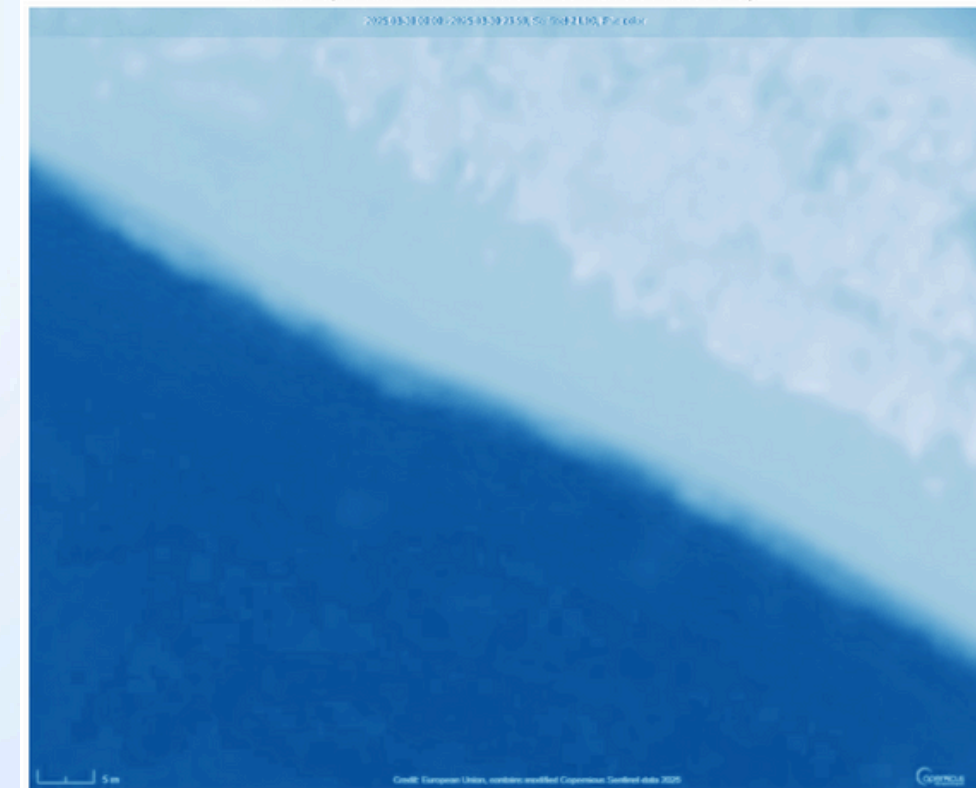
False Image



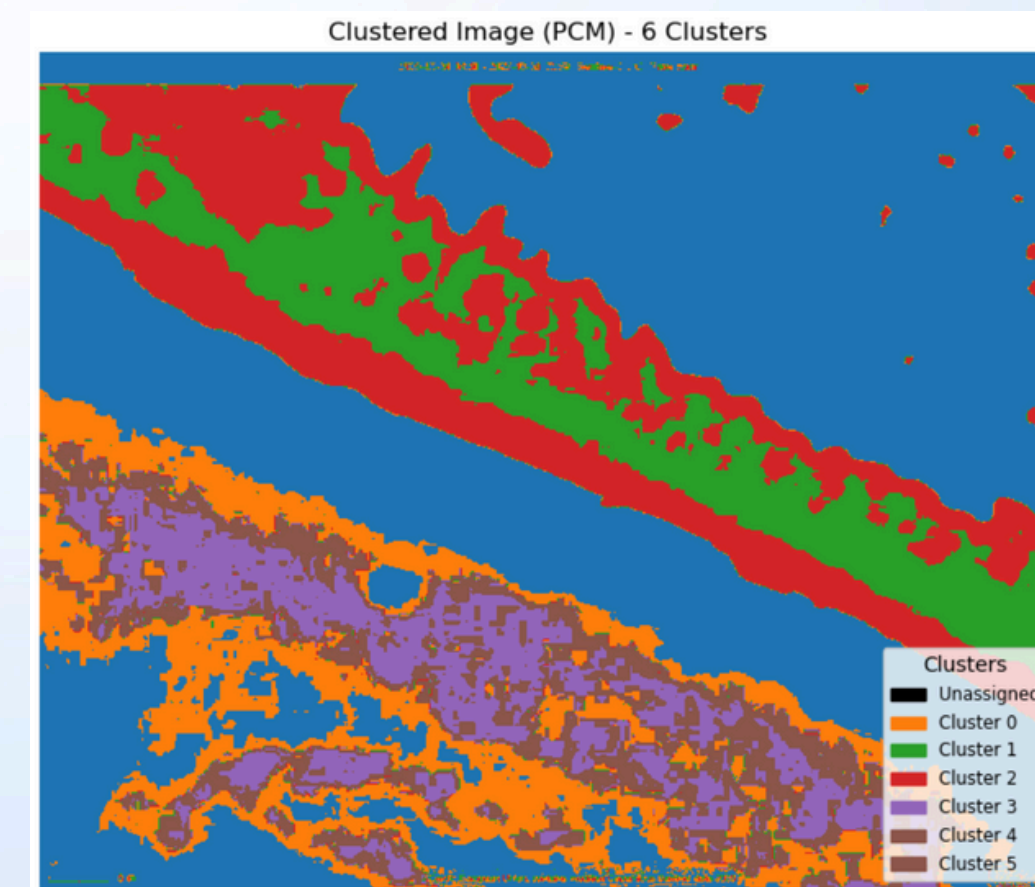
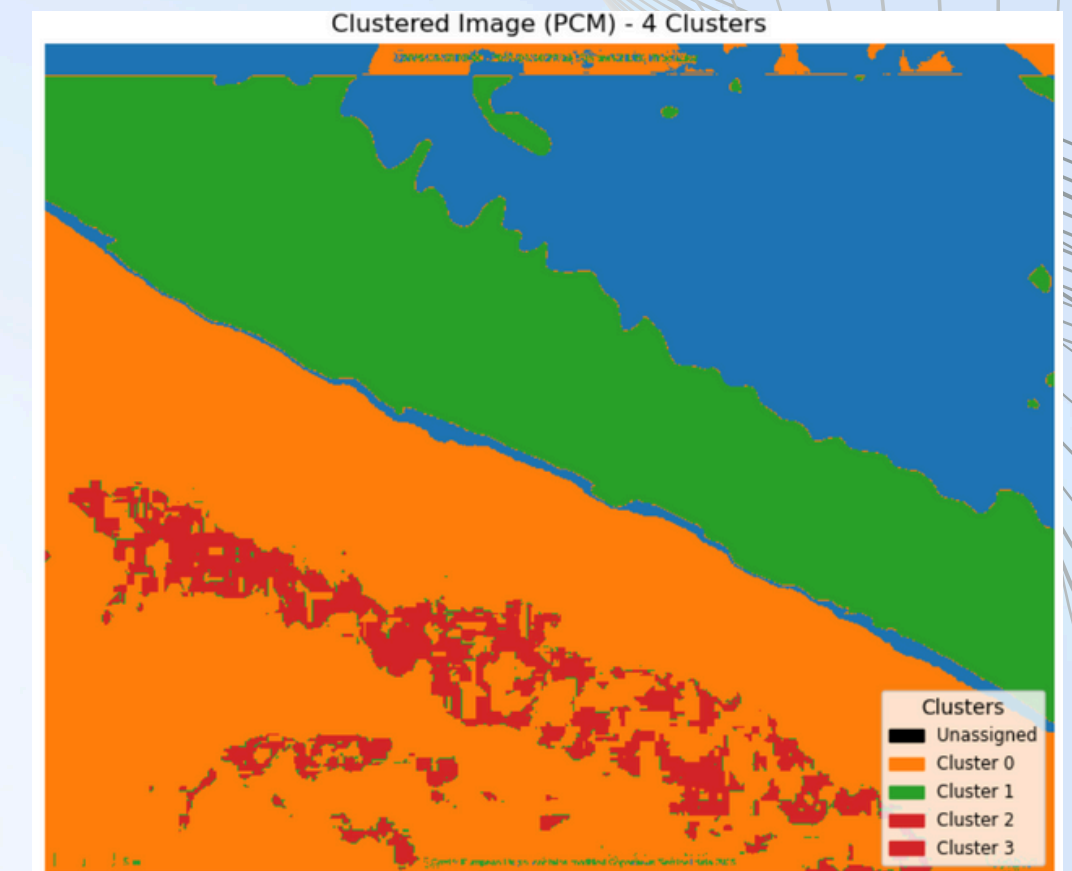
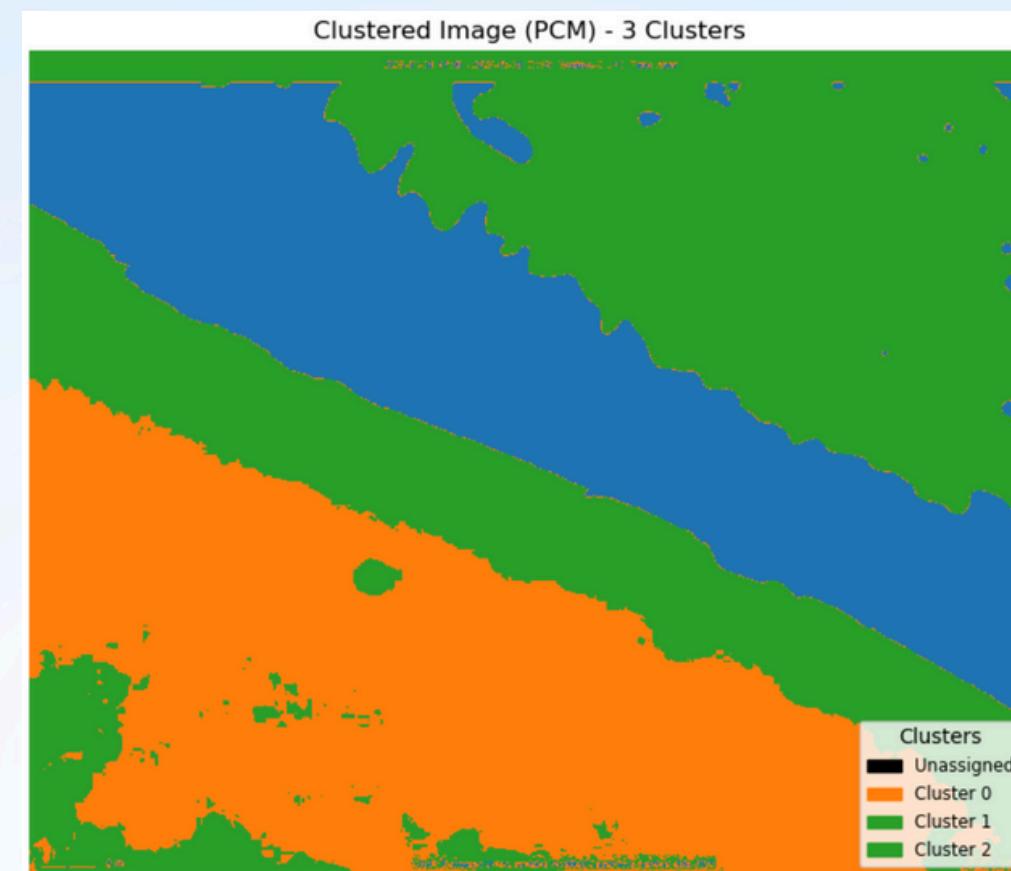
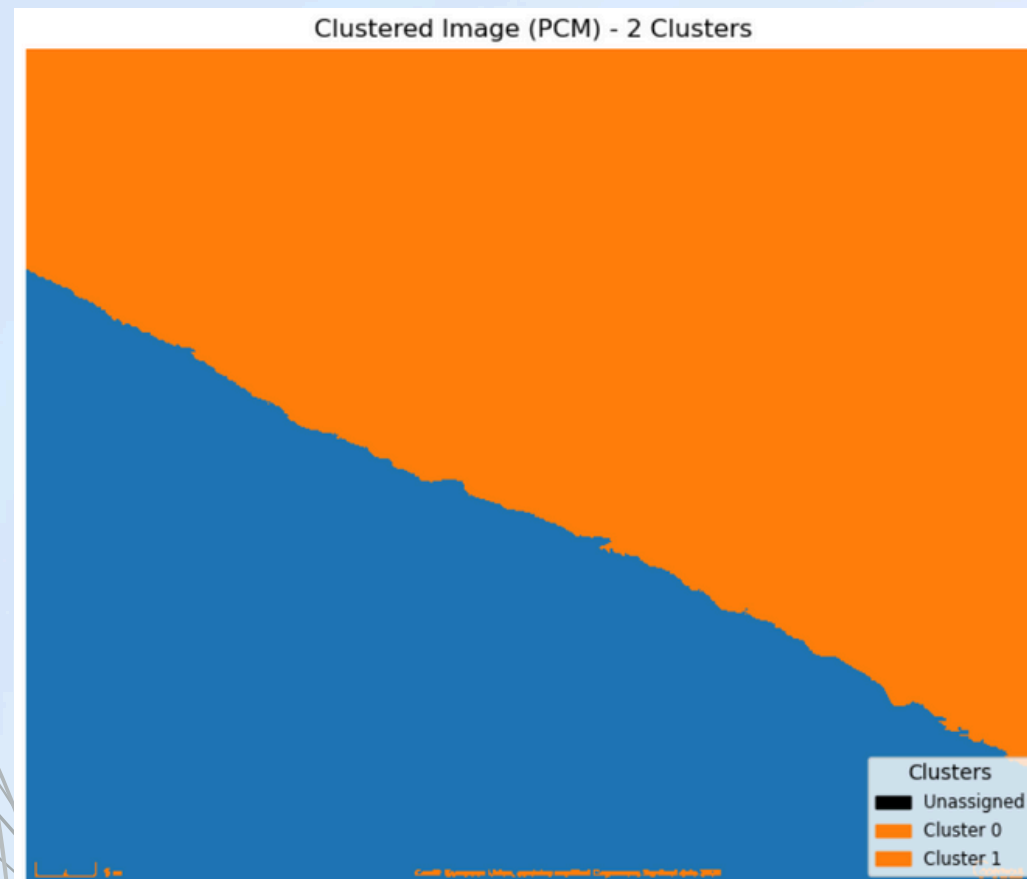
NDVI (Normalized Difference Vegetation Index)



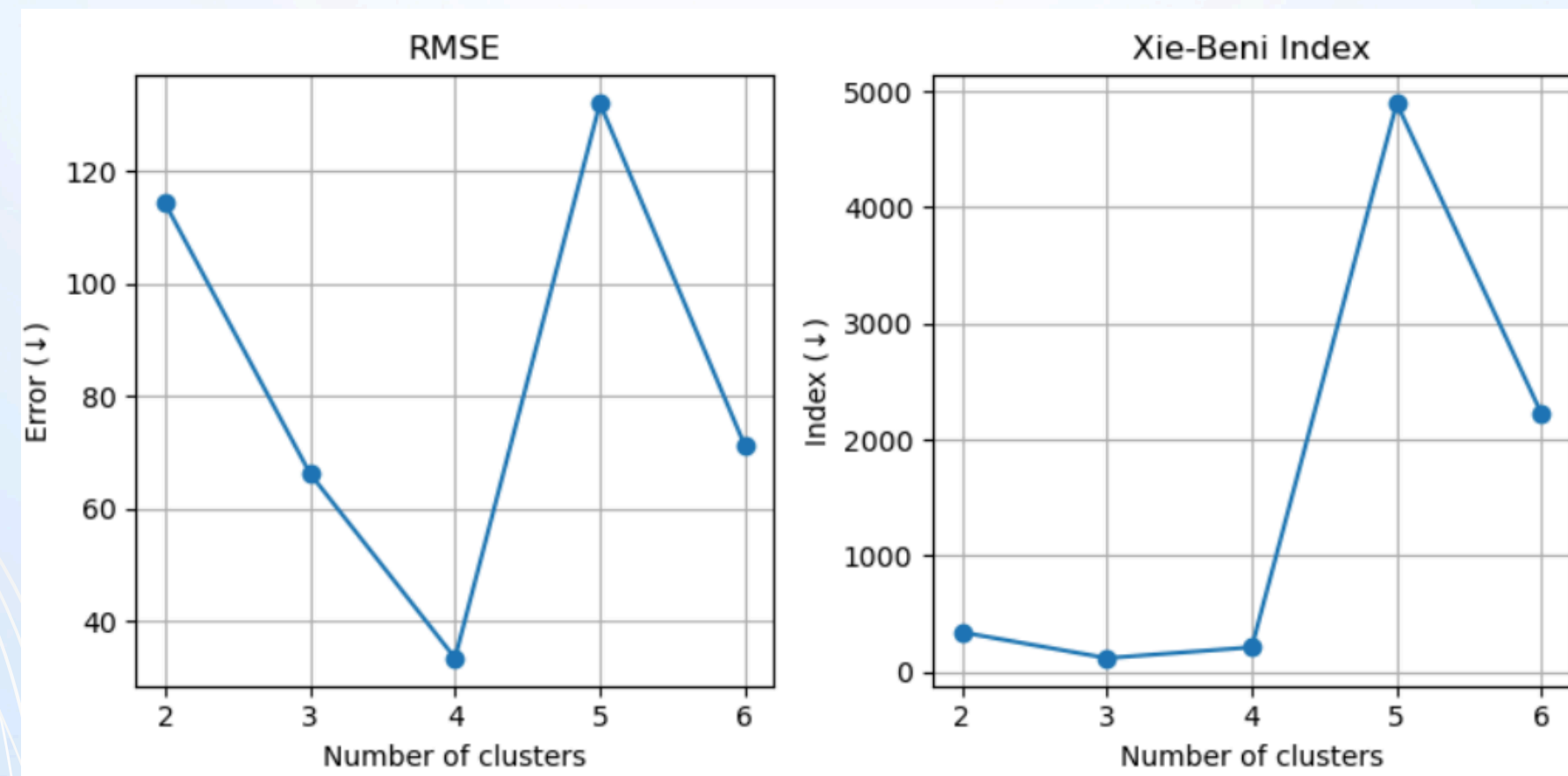
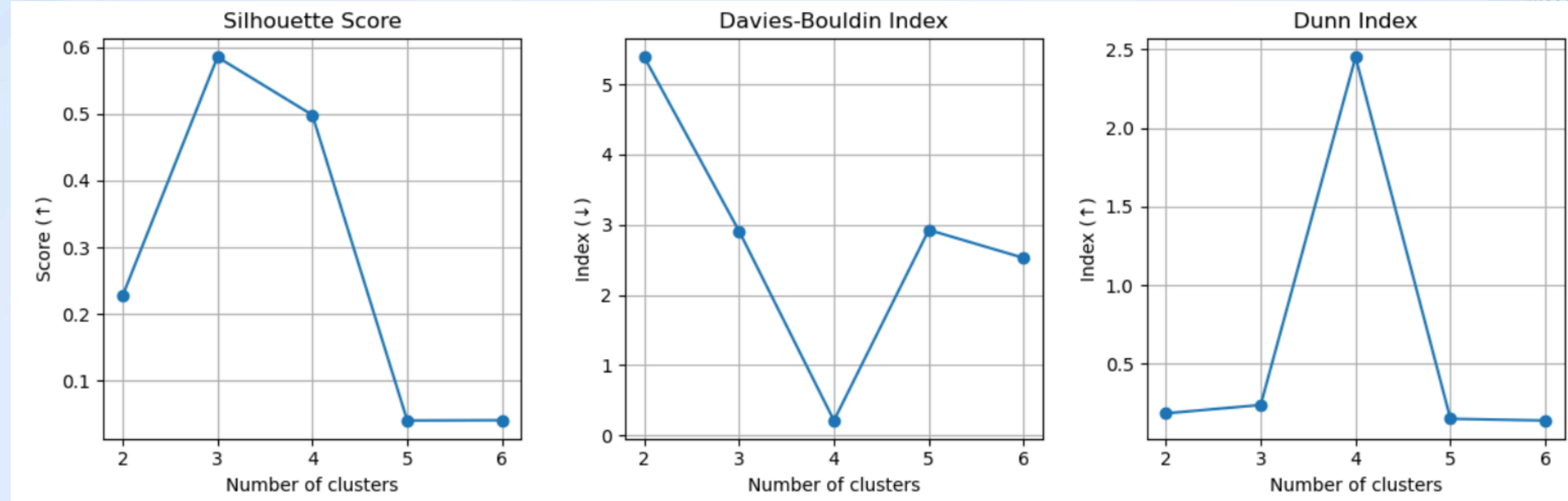
NDWI (Normalized Difference Water Index)



EXAMPLE - 1

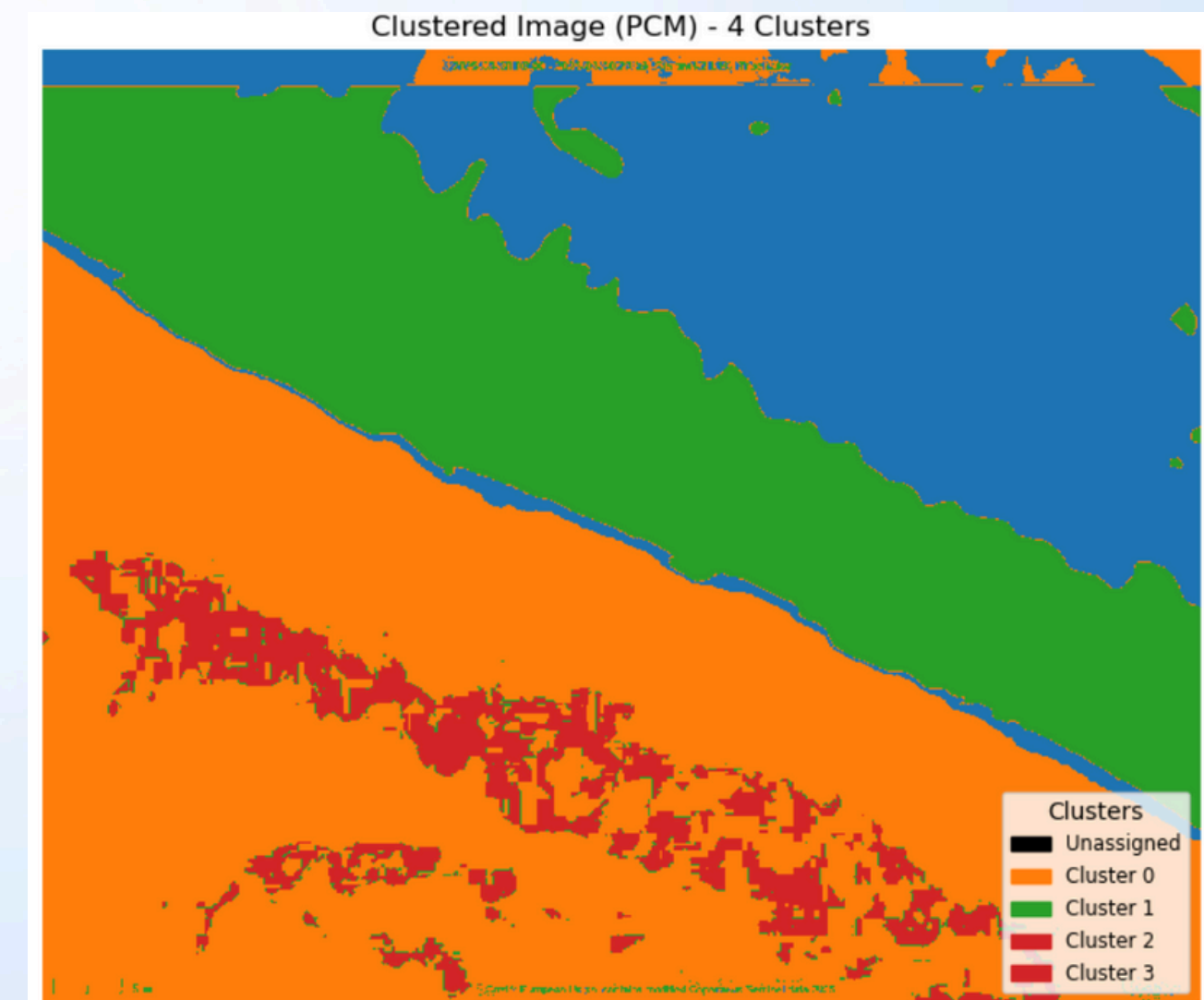
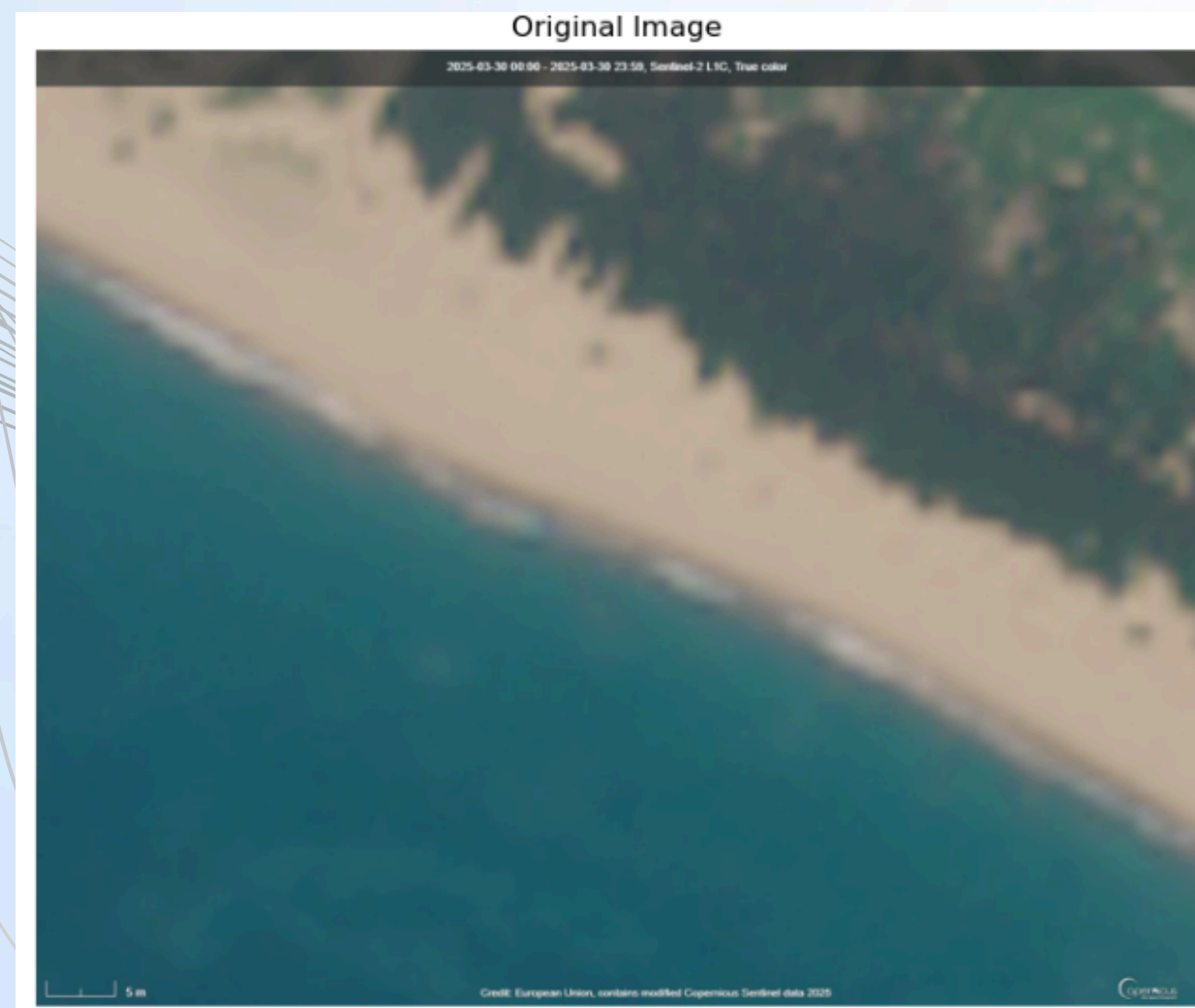


EXAMPLE - 1



EXAMPLE - 1

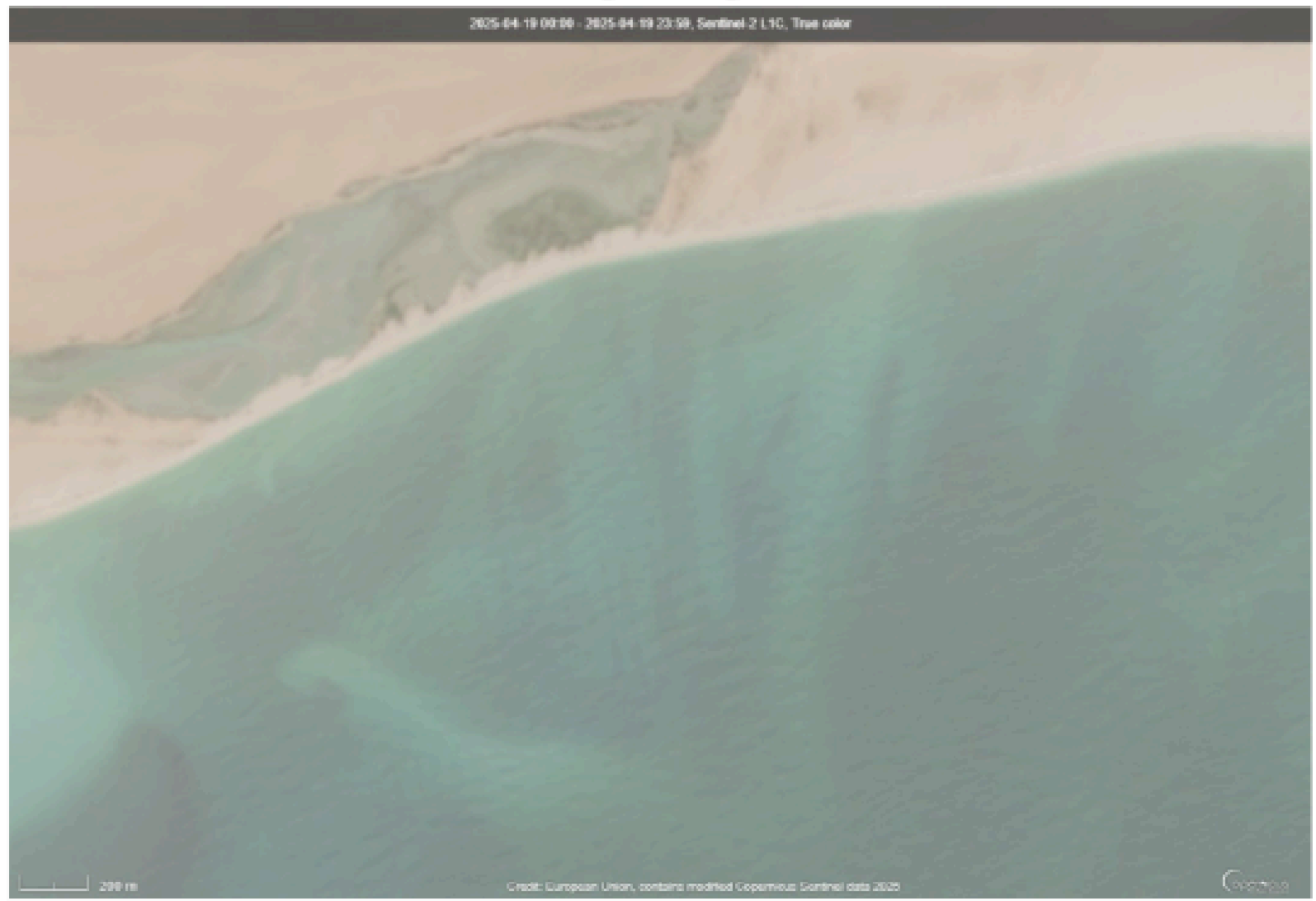
- From the results obtained in the previous slide, we see that the optimal number of clusters is 4.
- At this point, we see a large Dunn Index and a decently high silhouette score along with a small RMSE, XBI and DBI.
- All of these observations indicate that $C = 4$ is the optimal clustering.



RESULTS

EXAMPLE - 2

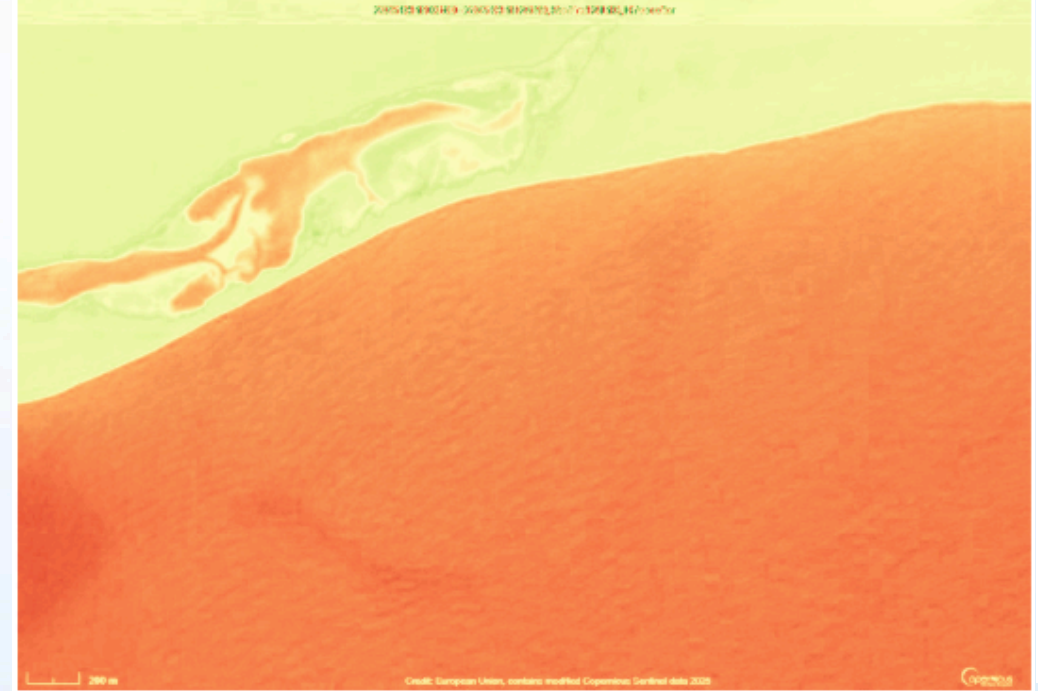
Original Image



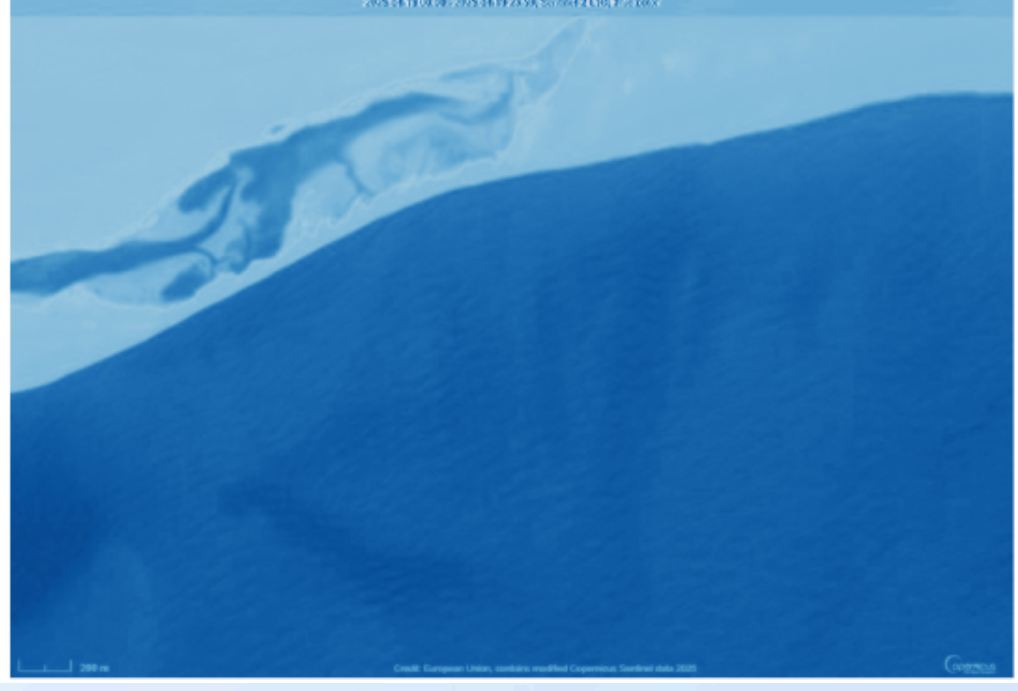
False Image



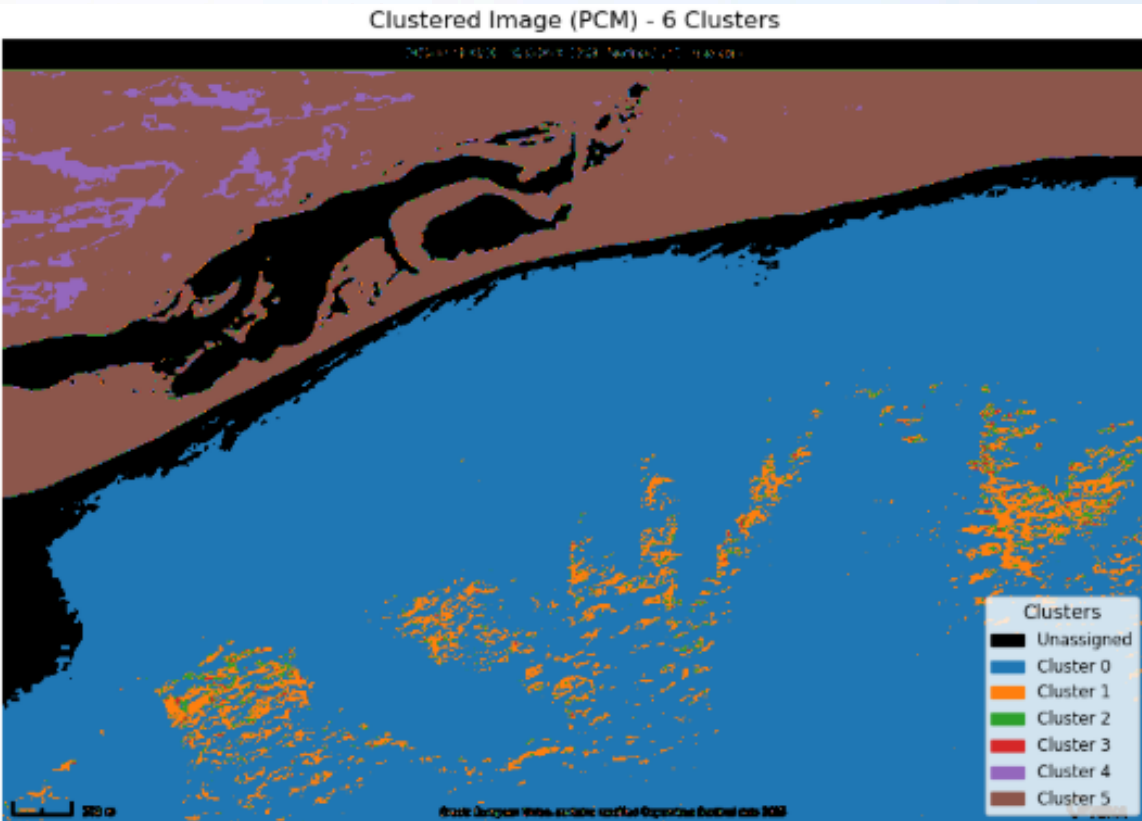
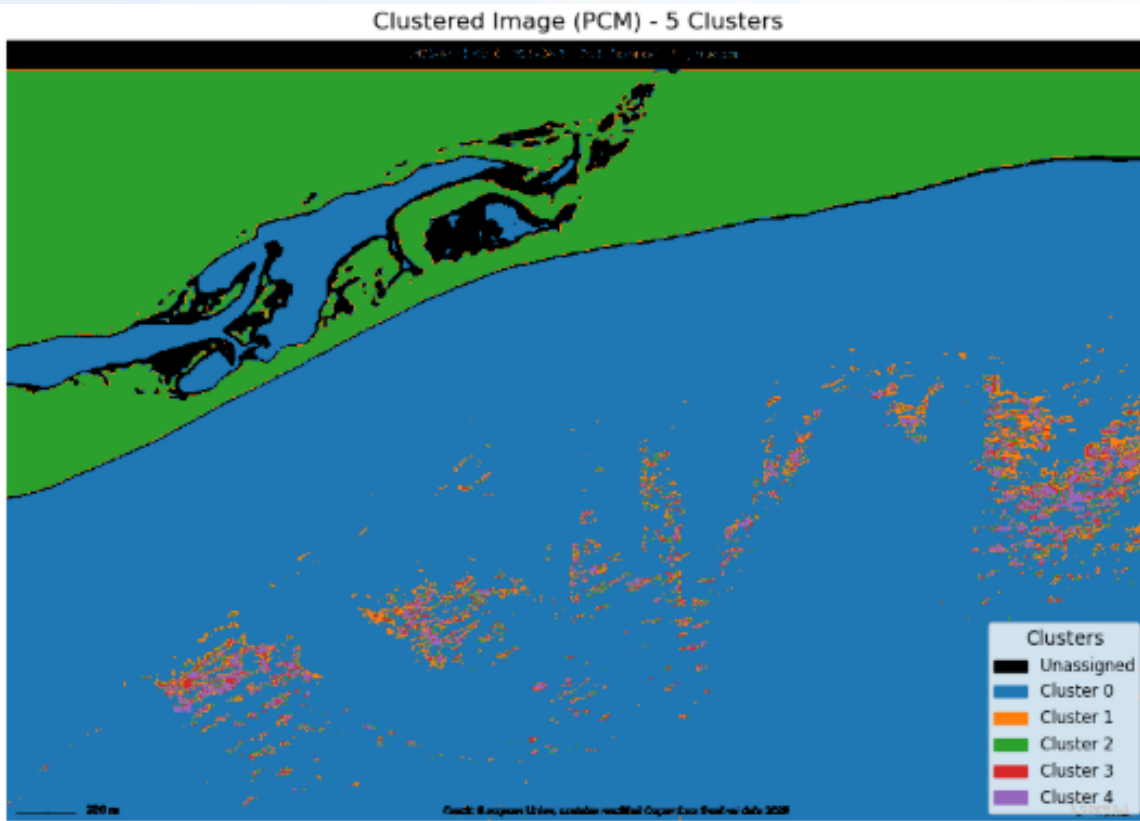
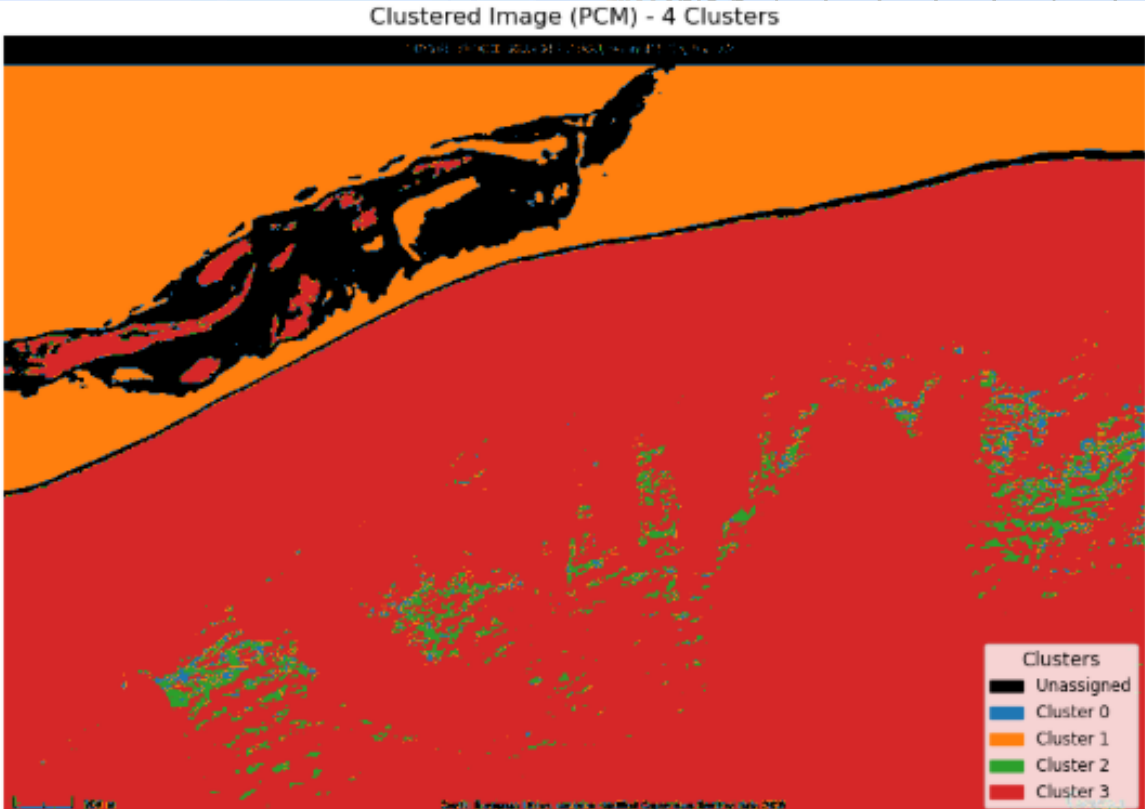
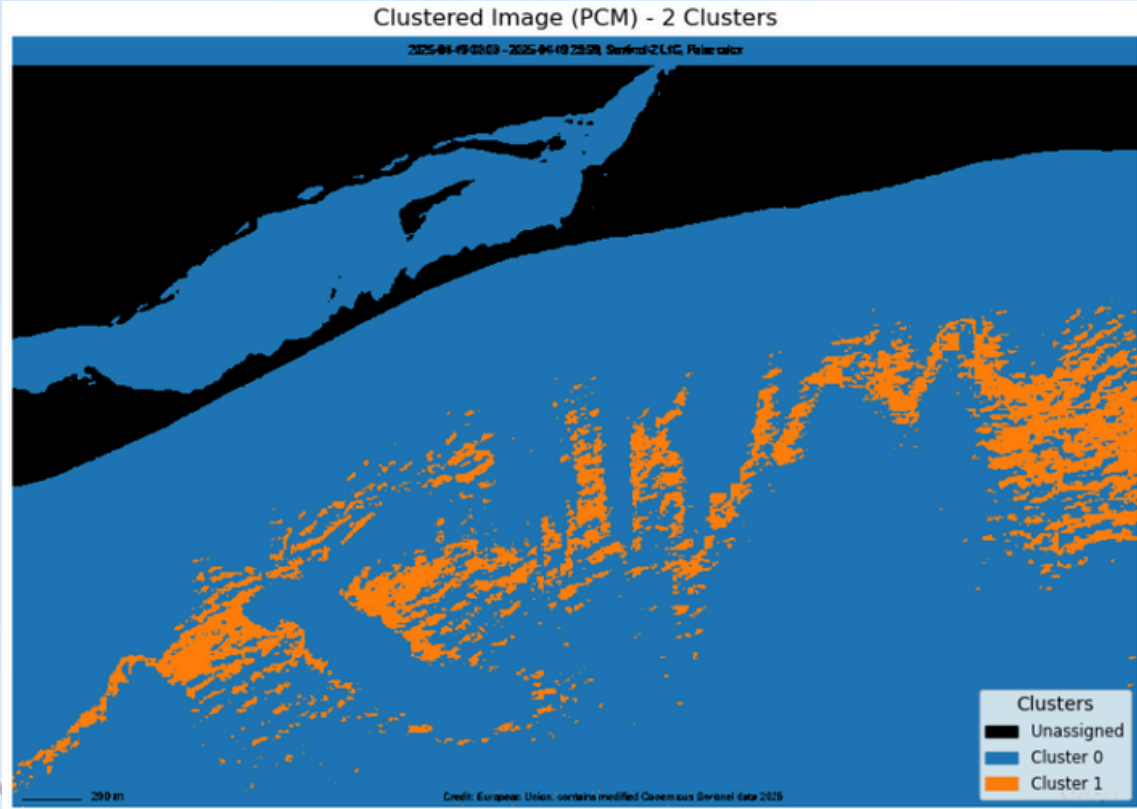
NDVI (Normalized Difference Vegetation Index)



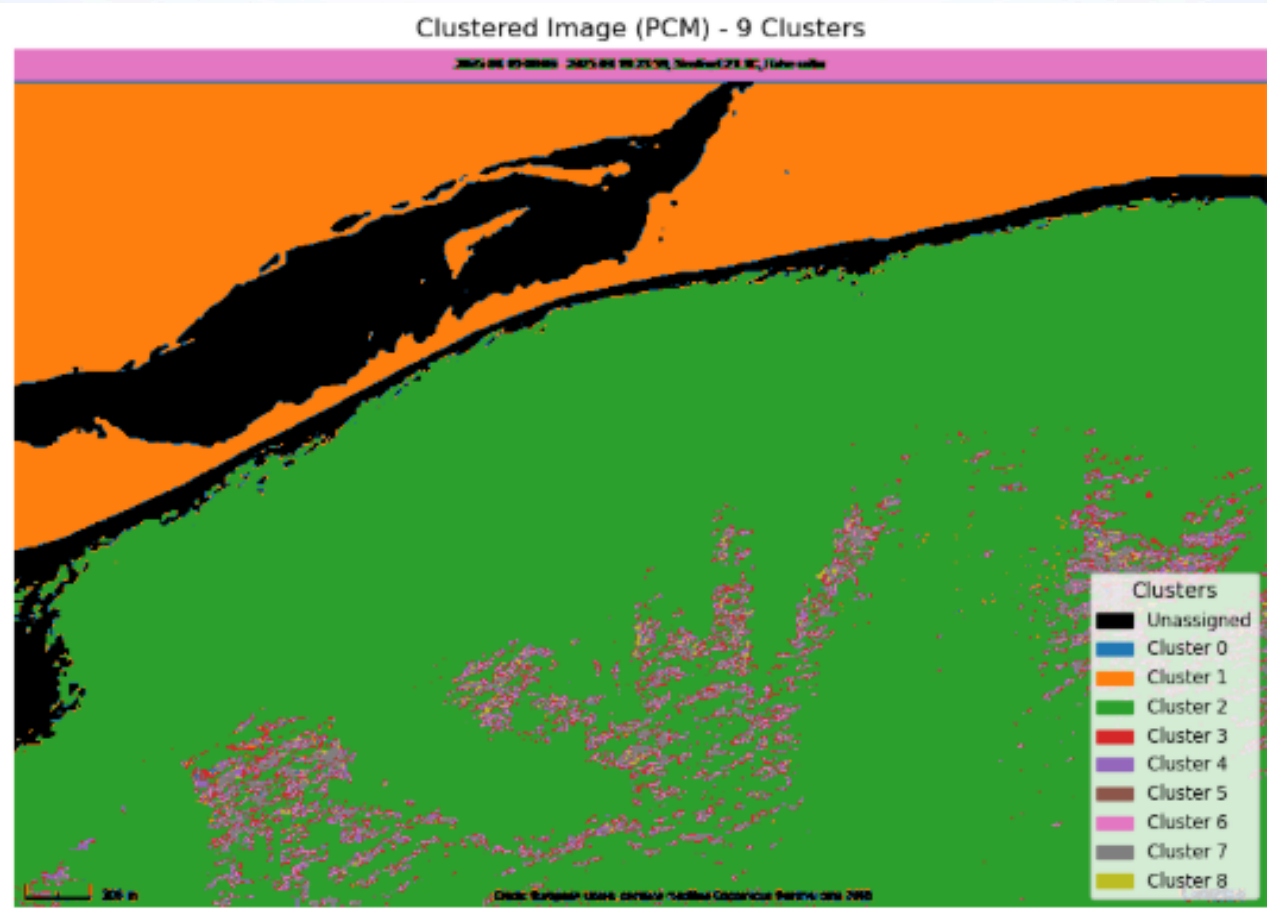
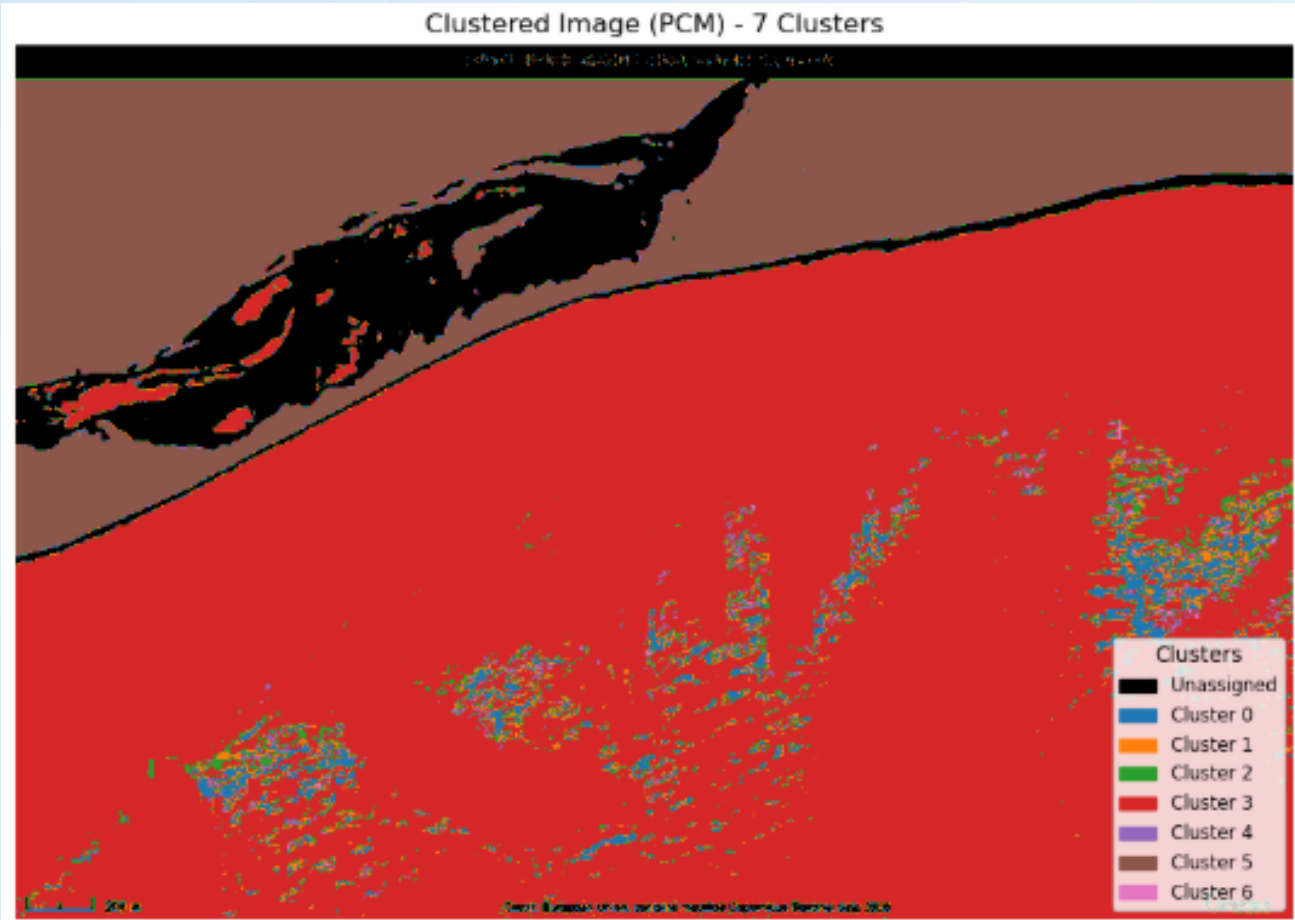
NDWI (Normalized Difference Water Index)



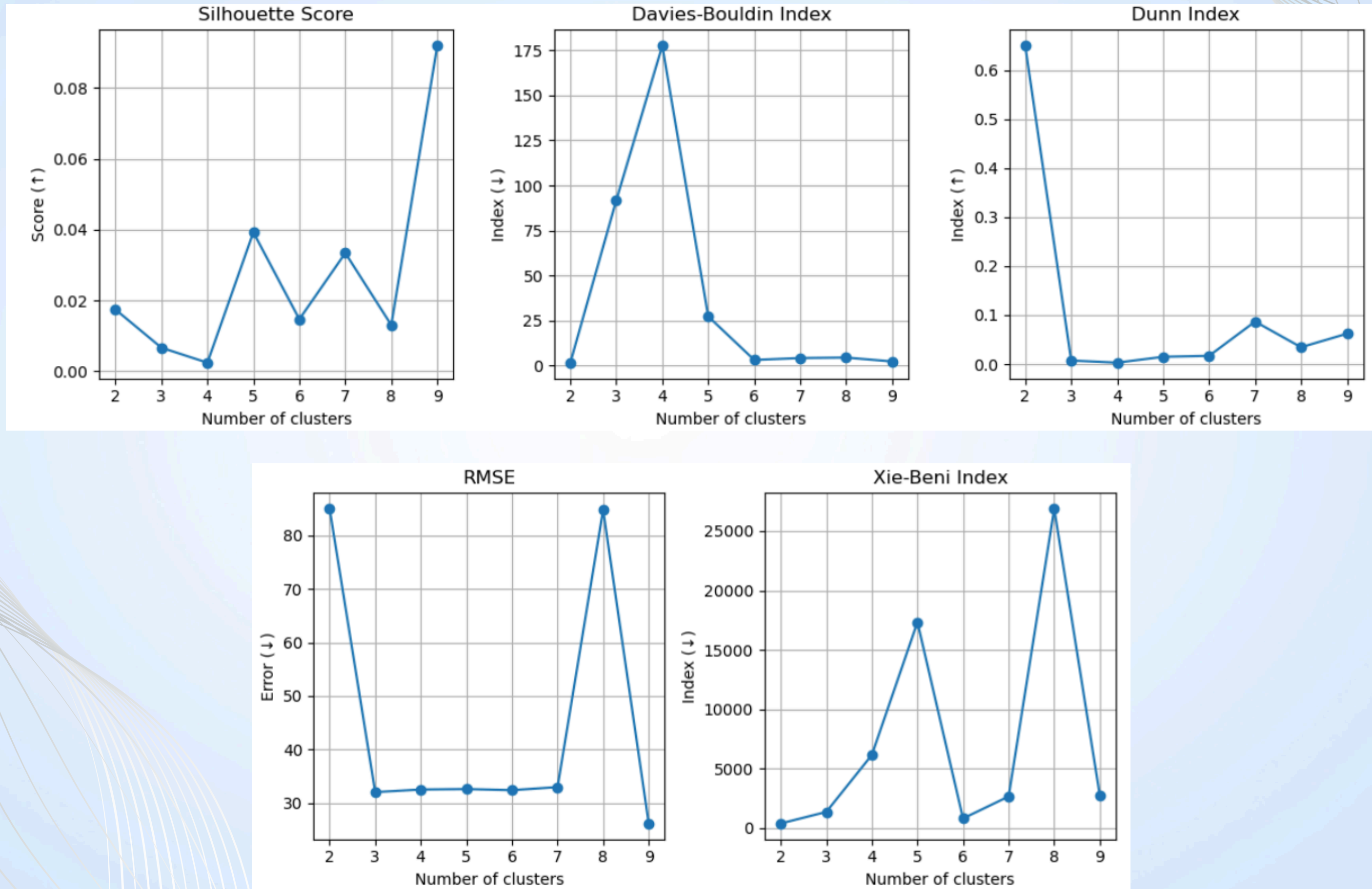
EXAMPLE - 2



EXAMPLE - 2

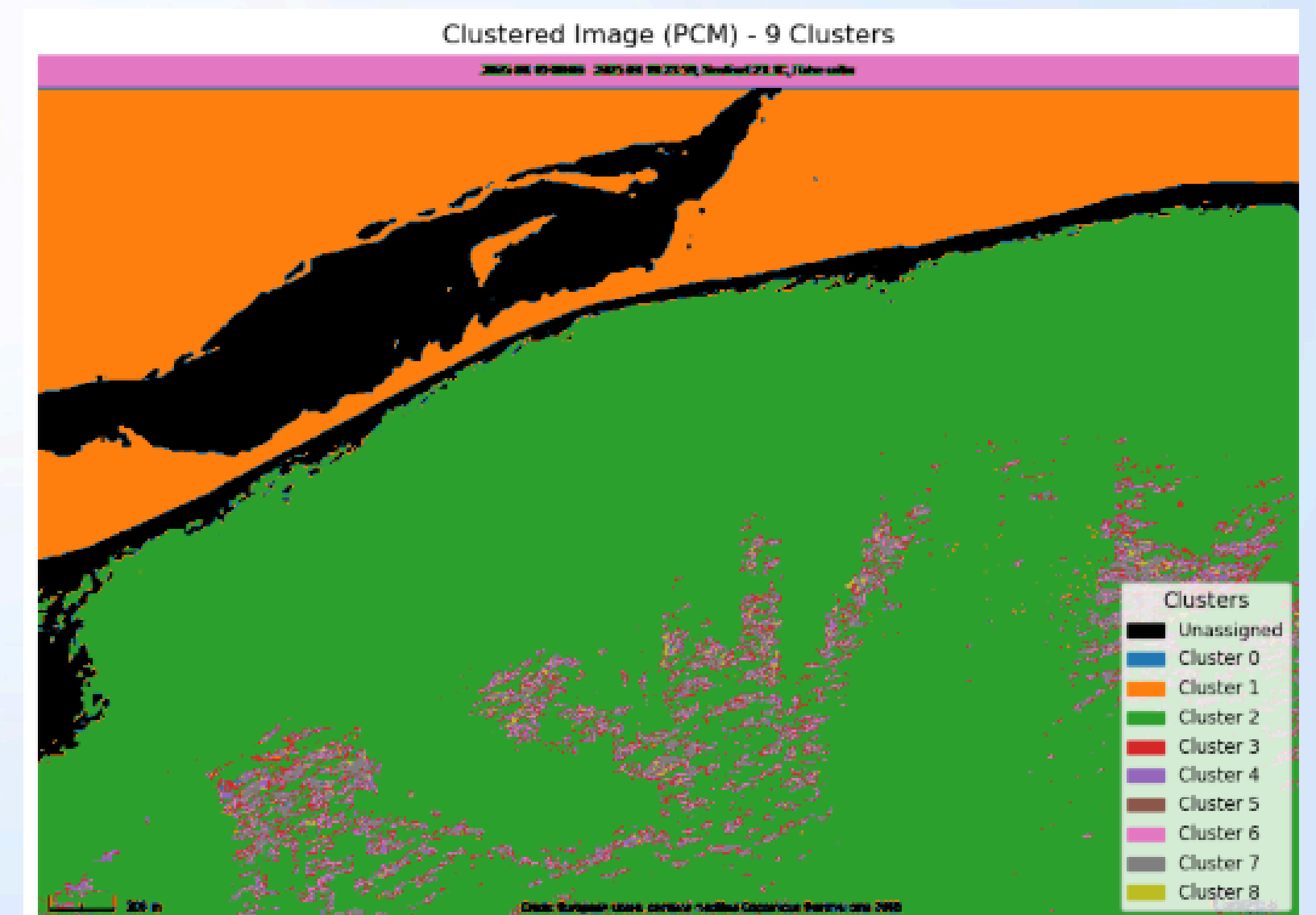
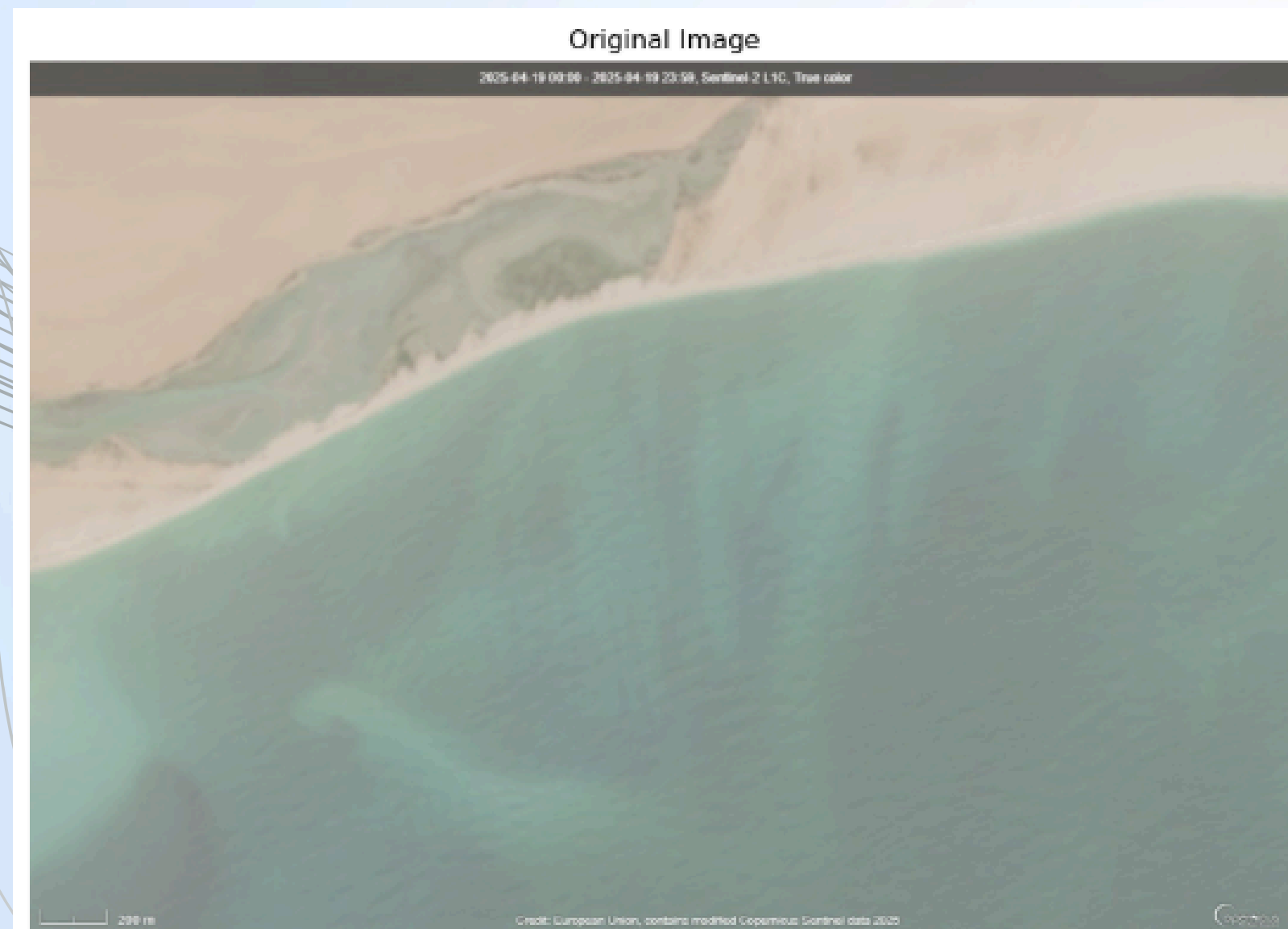


EXAMPLE - 2



EXAMPLE - 2

- In this case, we observe that the optimal number of clusters is 9.
- At this point, we see a minima in RMSE and DBI and a small XBI. There's a maxima in Silhouette score and a relatively high Dunn Index.
- All of these observations indicate that $C = 9$ is the optimal clustering.



RESULTS

EXAMPLE - 3

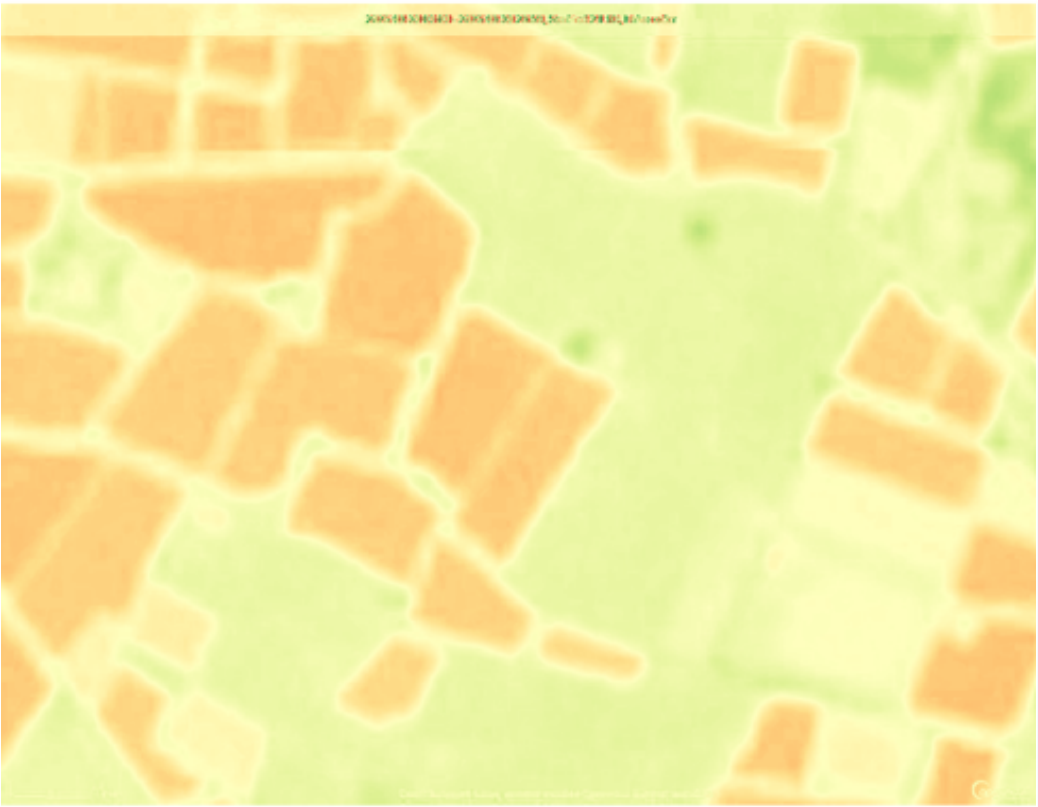
Original Image



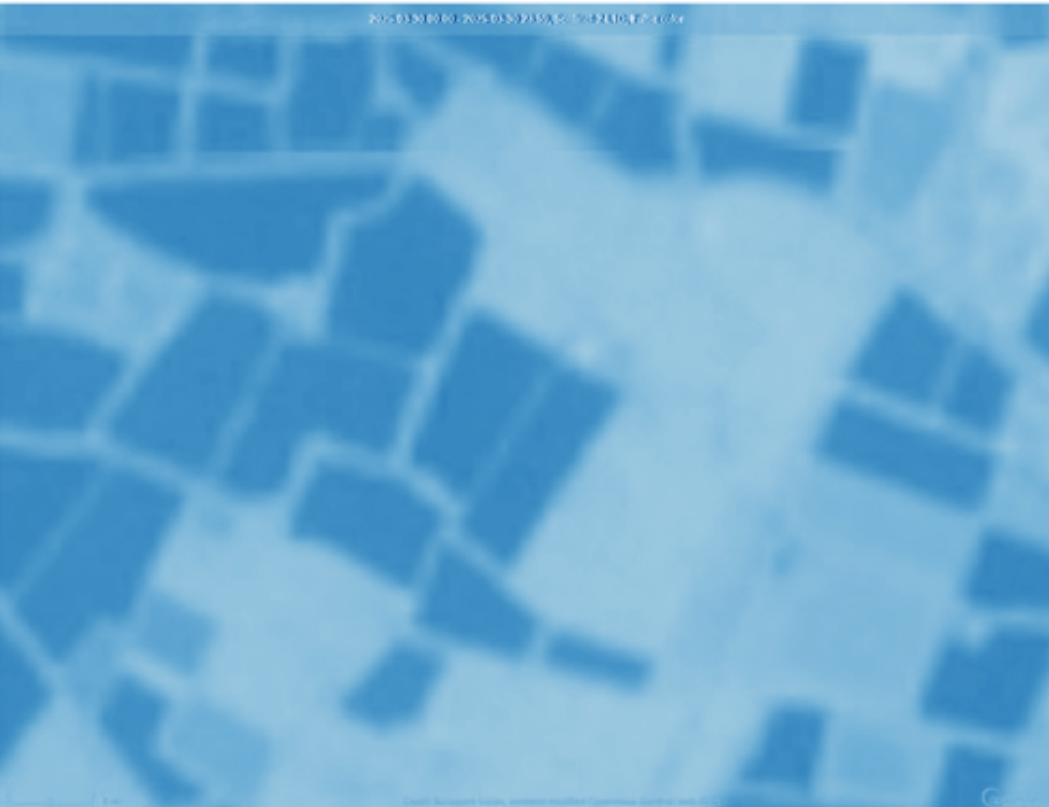
False Image



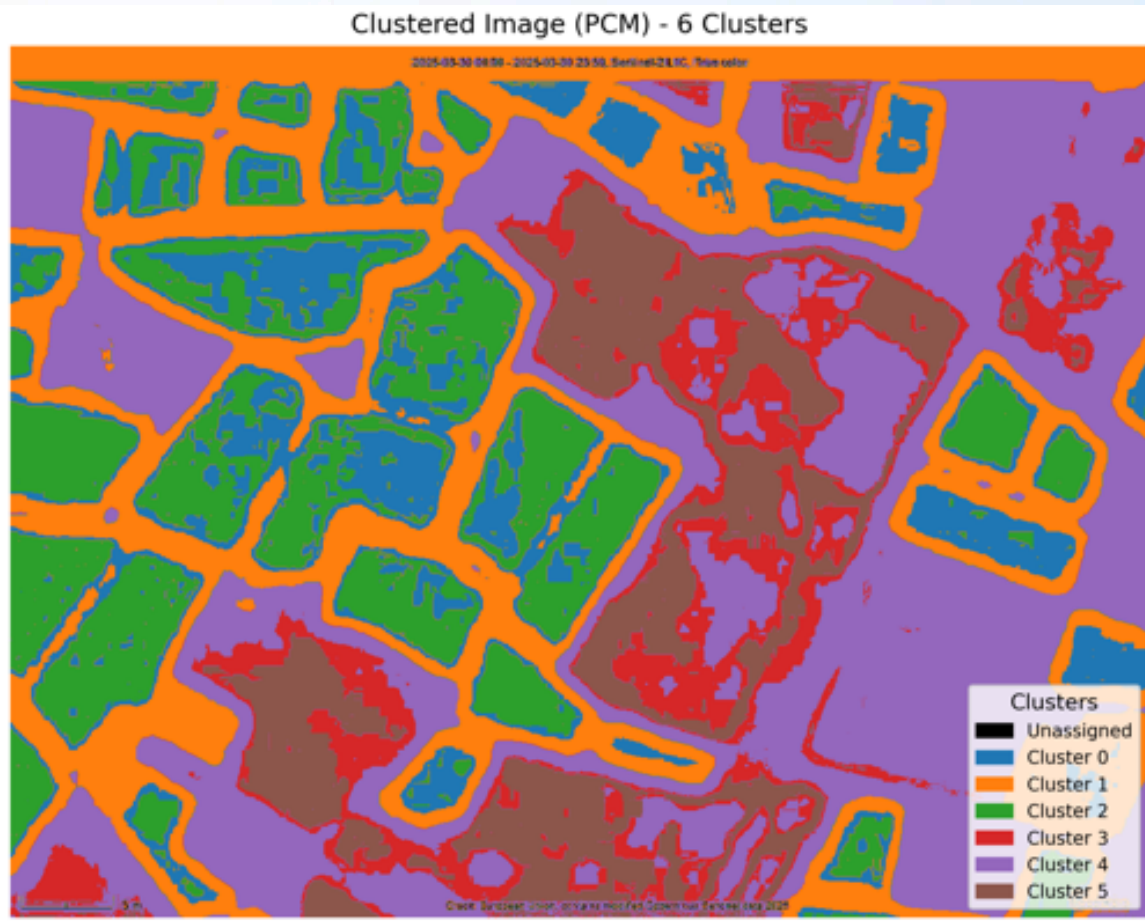
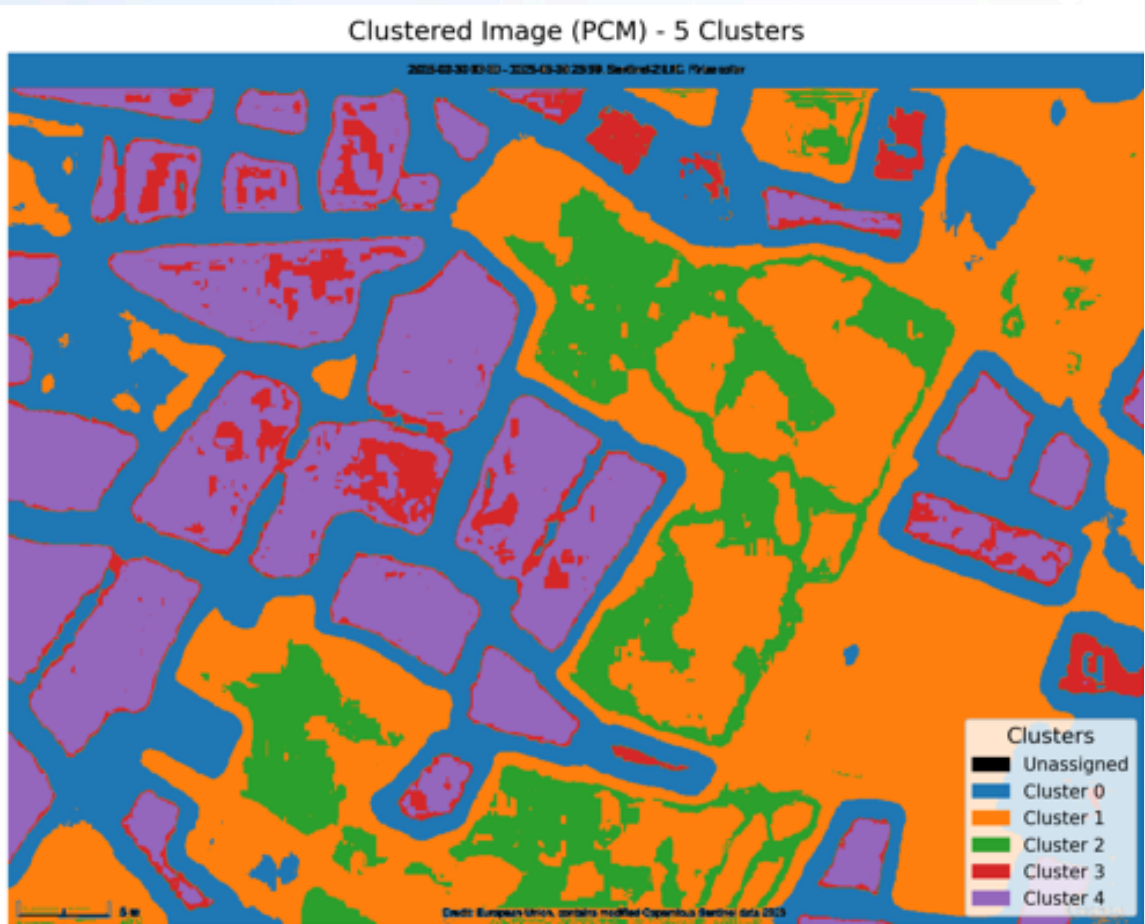
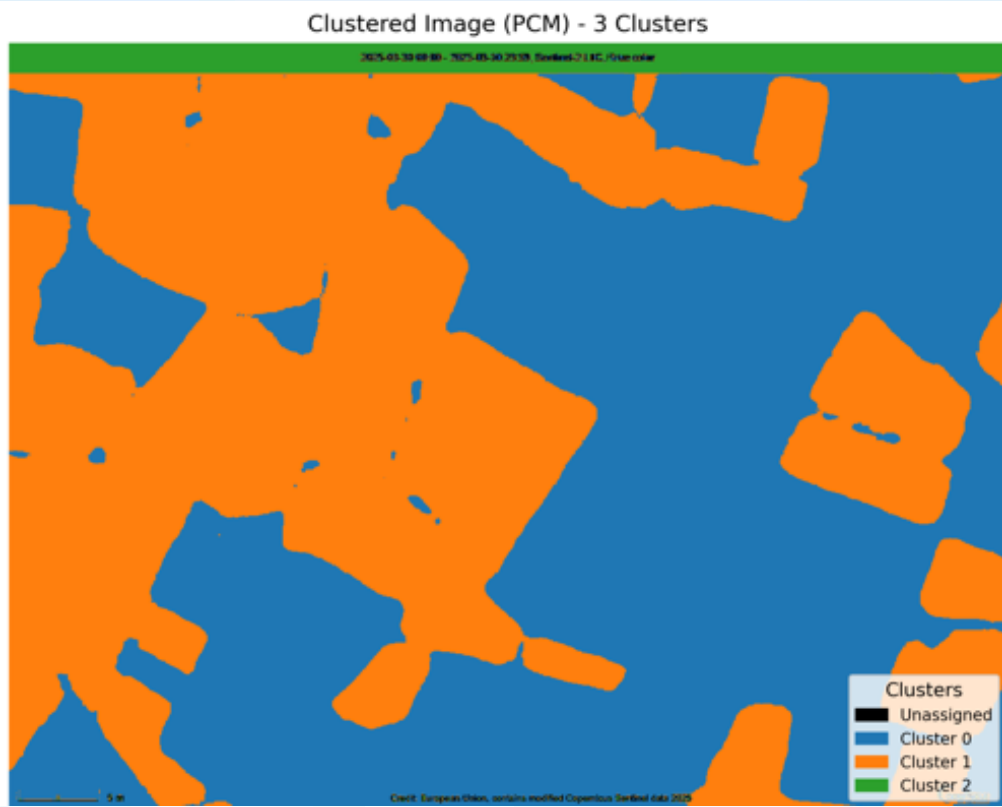
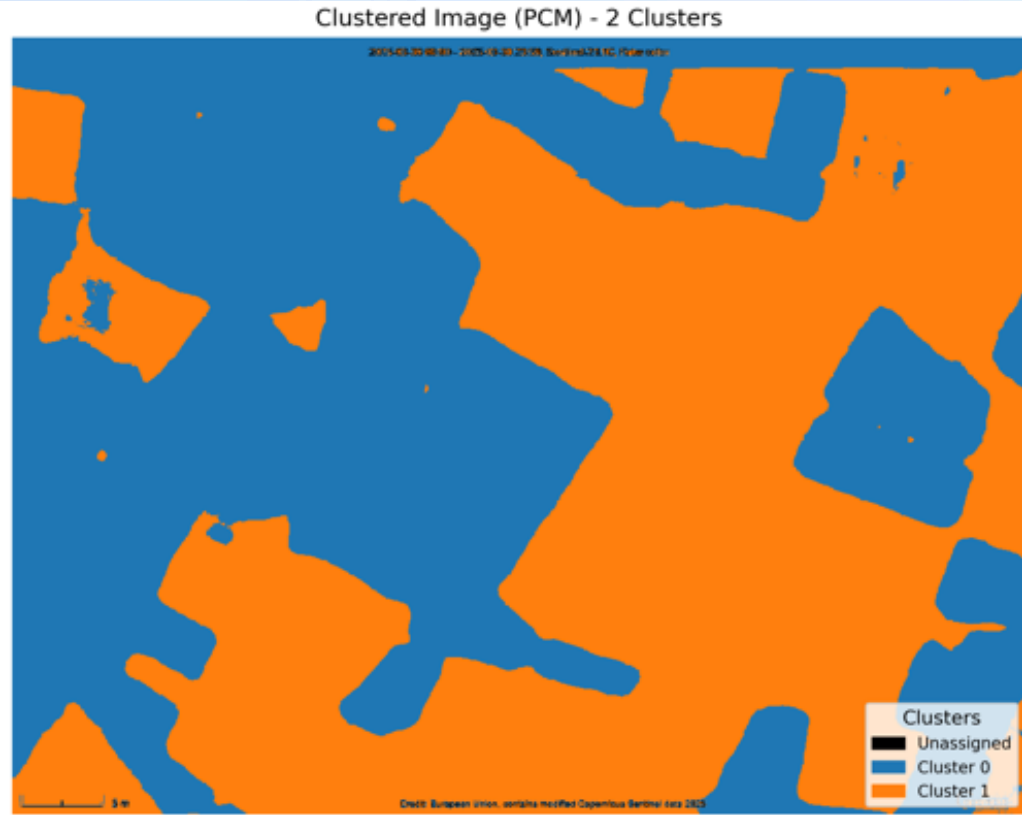
NDVI (Normalized Difference Vegetation Index)



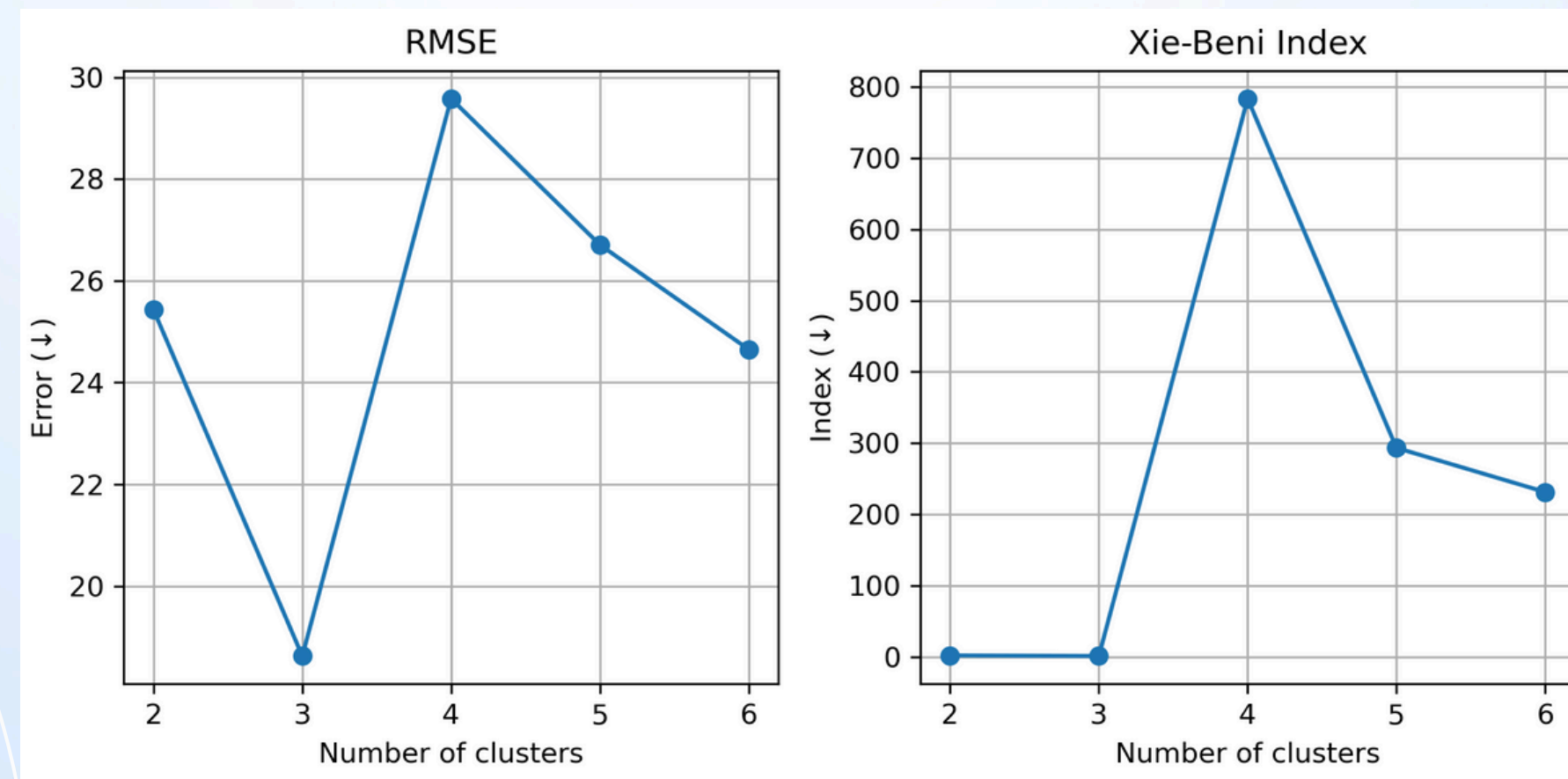
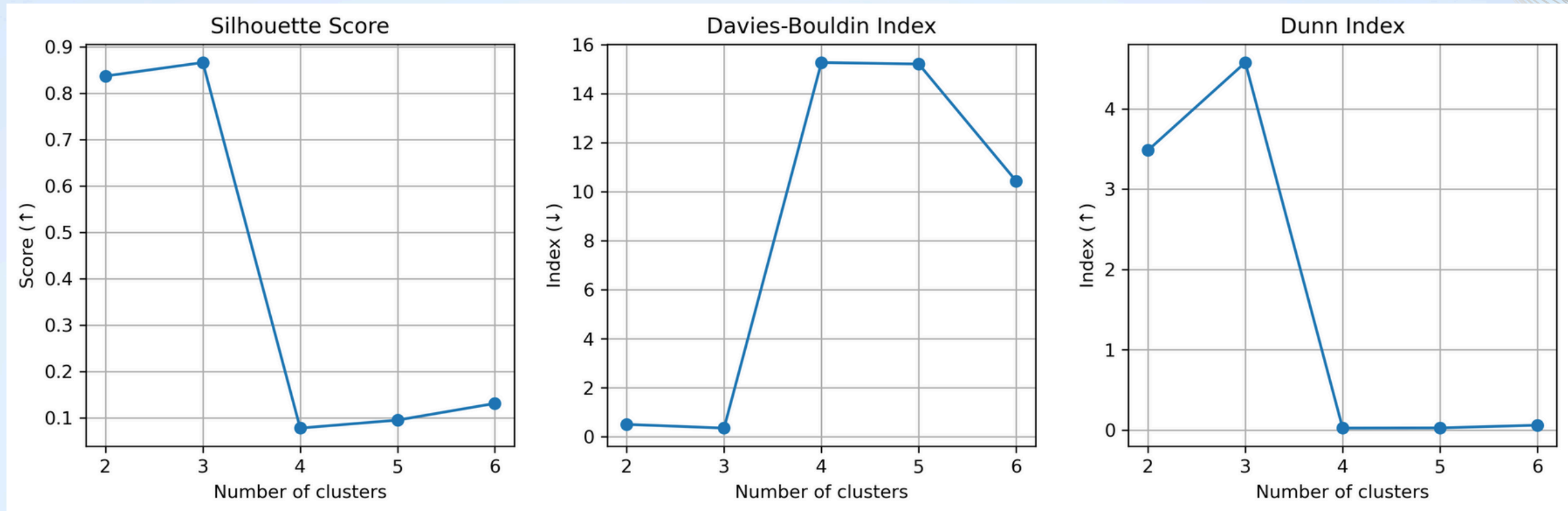
NDWI (Normalized Difference Water Index)



EXAMPLE - 3

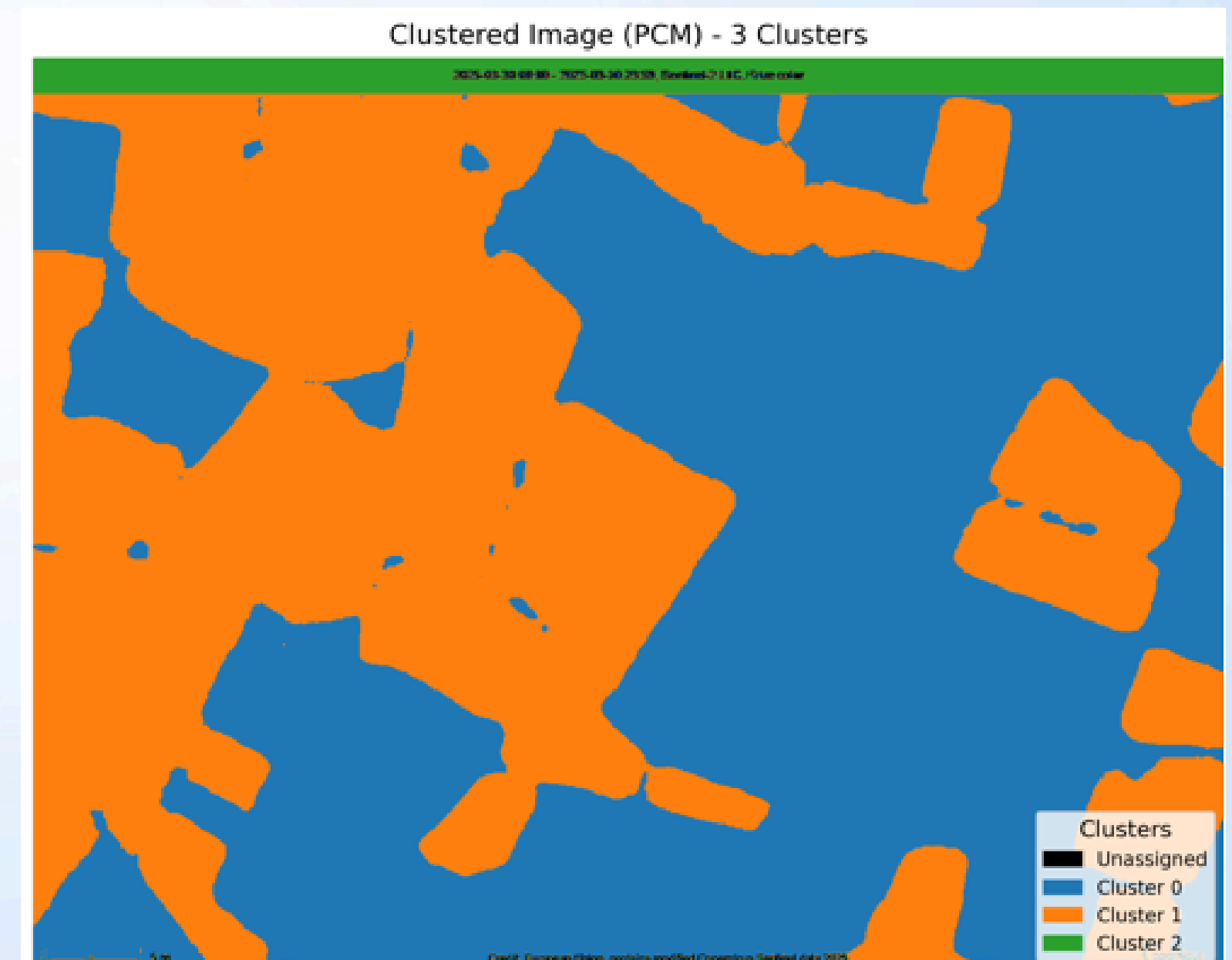
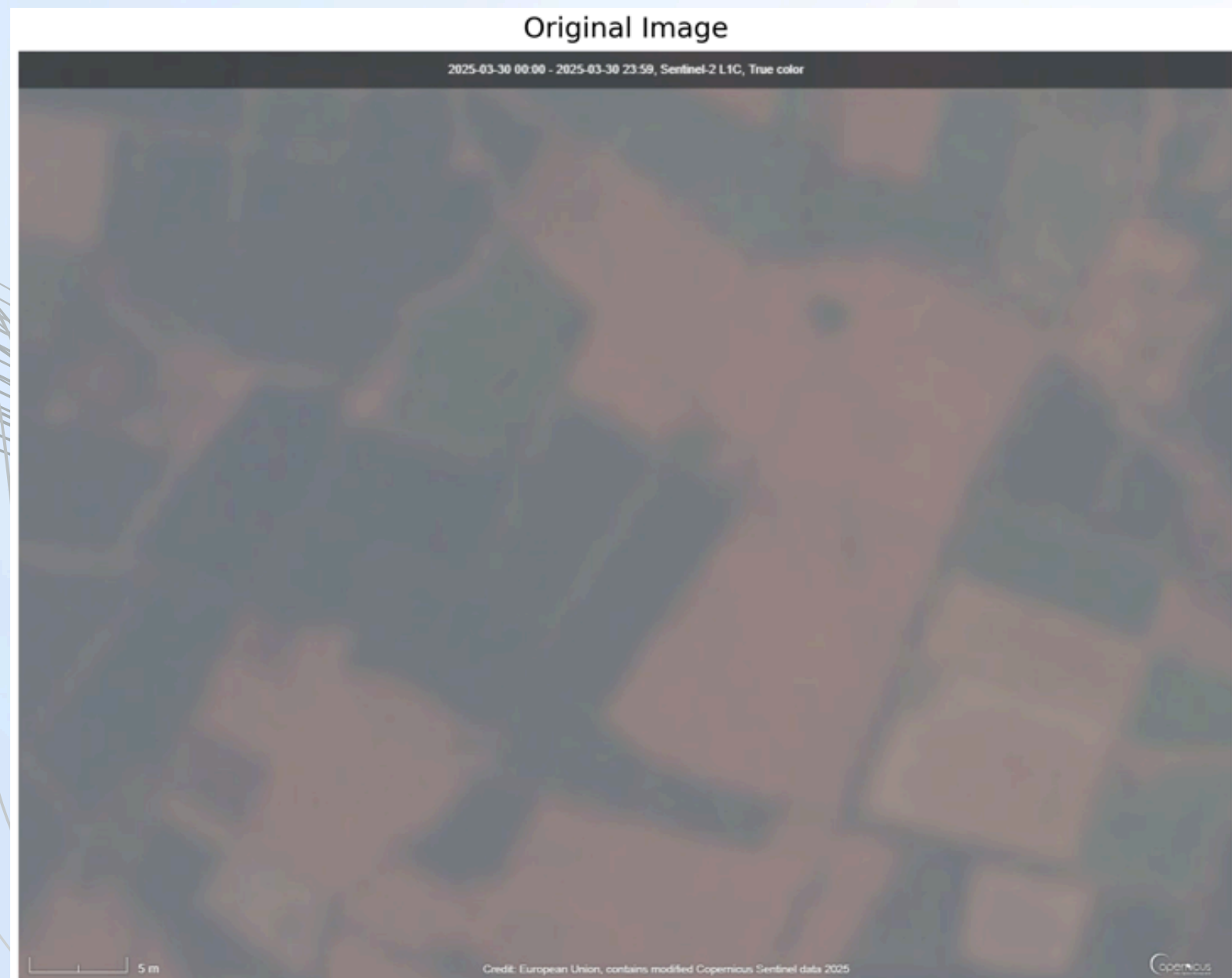


EXAMPLE - 3



EXAMPLE - 3

- In this case, we observe that the optimal number of clusters is 3.
- At this point, we see a maxima in Silhouette Score and Dunn Index and a small XBI, RMSE and DBI.
- All of these observations indicate that $C = 3$ is the optimal clustering.



The background features a series of concentric circles in shades of light blue, green, and yellow, centered behind the text. Additionally, there are several sets of thin, curved lines in grey and gold that flow from the corners towards the center, creating a sense of movement and depth.

THANK YOU